

WDW-9

Complete Joint Constraint-Kernel Realization Closure

A thirteen-fibre capstone for the constrained Wheeler–DeWitt realization

Amos Jay Maley

constraintsystems.institute.org | 5 August 2026

Exact scope. The theorem concerns the exact Branch-A, compact-hyperbolic, metric–real-scalar, profile-clock, source-attached, single-rate, no-spectator target accepted in WDW-1–8. It proves certificate-complete joint realization closure for that target. It does not assert a global clock, full-superspace or ultraviolet completion, singularity resolution, Standard-Model quantum-gravity closure, or uniqueness across changed targets.

Abstract

The Wheeler–DeWitt construction is often presented as a formal replacement of canonical momenta by functional derivatives. That formula does not by itself provide a common configuration carrier, proper-redescription descent, refoliation closure, a regulator and common domain, anomaly control, a nonzero joint solution, a positive physical quotient, a relational clock, or a covariant recovery theorem. WDW-1–8 construct those objects separately on a single source-attached Branch-A target. The remaining capstone problem is global compatibility: nonempty individual fibres need not possess a common section, and fibrewise uniqueness does not imply uniqueness of a complete realization.

This paper constructs the missing joint certificate. Thirteen target-presentation groupoids are fixed before candidate inspection. Their standing-bearing intersections are generated by twenty typed overlaps. The actual predecessor occupants are assembled into a tuple Ξ_9^* ; every overlap is checked in its inherited base type, including structured source ancestry, the complete graph domain, the function-valued profile clock, and the disjoint physical-anchor/recovery carrier. This proves $\Xi_9^* \in \mathfrak{J}_9$ and hence positive population of the thirteen-fibre inverse limit.

The primary interfaces are then completed in the value-reflecting slice $\mathbf{Set}^{\widehat{S}_9}/\widehat{W}_9$, with $\widehat{W}_9(b_e) = \mathfrak{B}_e$ at every overlap base. Their explicit tensor colimit L_9^0 is kept distinct from the complete certificate $\mathfrak{C}_9^0$, which carries the thirteen-tuple, its ledger, and its universal cocone. The adjunction first induces a mediator from each ledger; colimit functoriality then defines the tensor probe through the mediator-isomorphism groupoid. The fibre and incidence probes assemble through actual certificate projections into Θ_9^{cert} . Its kernel pair is the maximal certificate skin, while each fibre has its own augmented-probe kernel pair. Fixed-domain and same-scope exhaustion classify a sorted candidate grammar. Antecedent fidelity fixes no augmented-probe output; broad testable fibre domains are formed before thirteen candidate-specific response-realization relations test the equations, universal properties and commuting diagrams actually carried by each response. Only then do the independent predecessor uniqueness theorems act. Skin is a binary relation rather than a competing status, and no kernel-pair equality is presumed to construct a presentation arrow. After compatibility is proved to descend to fibre classes, those non-output-loaded finality theorems leave a live residue of thirteen classes, one compatible assignment, and one lawful orbit.

The resulting physical object is the already accepted WDW-6 nonzero joint kernel and positive rank-one quotient, not a WKB comparison state. Bulk coordinate time, lapse, shift, and foliation are presentation coordinates when the complete probe is fixed; the local profile clock and nonconstant complete record remain standing-bearing. Boundary time is absent on the compact boundaryless target rather than universally denied. The Wheeler–DeWitt equation is consequently placed as a downstream constraint-native realization of the common covariant, canonical, quantum, clock, and recovery architecture. The theorem is certificate-complete for the one frozen target and fail-closed at every inherited or newly identified boundary. After that core closure is proved, the positive rank-one quotient, projective state class, physical form, represented algebra, relational continuation, record, and failure boundary are assembled into an explicit quantum-standing export. The export supplies an occupied realization of the structural role used by downstream quantum dynamics, while expressly withholding a flat spatial carrier, $L^2(\mathbb{R}^d)$, Δ , $\text{SSQ}_1(H)$, V , m , and $\hbar$.

Principal result

Theorem 0.1 (Kernel-faithful joint constraint-kernel realization closure). *Let $\mathcal{D}_{\text{WDW}}^{\text{core}}$ be the exact Branch-A target fixed in Section 1. Assume the accepted WDW-1–8 and AASC interfaces listed in Table 1; no further analytic or physical premise is understood. Then:*

1. *the actual thirteen predecessor occupants form a compatible, nondegenerate joint section*

$$\Xi_9^* \in \mathfrak{J}_9, \quad \mathfrak{J}_9 \neq \emptyset;$$

2. *the completed witness-reflecting tensor ledger is the colimit L_9^0 of the declared generator/overlap diagram; the complete certificate is separately populated by*

$$c_9^* = (\Xi_9^*, L_9^0, \kappa^*) \in \mathfrak{C}_9^0,$$

and the lawful certificate and fibre skins are exactly

$$S_9^{\text{cert}} = \text{KerPair}(\Theta_9^{\text{cert}}), \quad S_{9,i} = \text{KerPair}(\widehat{\Theta}_{9,i});$$

3. *every same-target generated candidate has one complete status, the thirteen target-relative fibres each have one lawful residue class, and the live residue formed from all compatible injections satisfies*

$$\left| \text{Residue}_9^{\text{live}} \right| = 13;$$

4. the complete assignment space has one certificate-lawful orbit,

$$|\Sigma_9/\Gamma_9^{\text{law}}| = 1;$$

5. the physical carrier in the joint certificate is the accepted nonzero WDW-6 joint constraint kernel and positive quotient

$$\mathcal{H}_{\text{phys},9} \cong \overline{\mathcal{E}_A^{\text{ren}}/\text{Rad } B_A} \cong \mathbb{C}, \quad \mathfrak{A}_{\text{phys},9} = \mathbb{C}I;$$

6. bulk coordinate-time data carry no independent standing under the complete probe, while the certified local profile clock and nonconstant complete record do; boundary/asymptotic time is absent from, rather than prohibited beyond, the core target; and

7. the Wheeler–DeWitt equation occupies the downstream constraint-native lineage established in Theorem 8.3. It is not assumed as a primitive momentum-substitution rule and does not acquire conclusions outside the exact failure boundary of Theorem 9.1; and

8. downstream of the core endpoint, the occupied physical line and its typed structures form the quantum-standing export Ω_9^{WDW} of Theorem 10.4, without being promoted to a spatial Schrödinger carrier.

Consequently every gate G9-01–G9-18 is satisfied and

$$G_9 = \text{PASS}, \quad H_9^{\text{QS-EXPORT}} = \text{PASS}.$$

The core proof is the ordered chain W9.1–W9.18 in Sections 1–9; the strictly downstream export is W9.19 in Section 10. Positive joint population precedes completion, probe construction, fibre finality, live-residue formation, and orbit counting. No conclusion of Theorem 0.1 is used to populate an antecedent.

Exact reading. “Complete” means complete relative to one frozen certificate and its declared comparison grammar. “Unique” means one orbit after quotienting the maximal congruence invisible to every registered standing-bearing probe. A changed source, boundary, operator grammar, ordering, counterterm, anomaly class, clock, recovery carrier, or report coordinate is a different target instance, not a suppressed rival.

Contents

Principal result	i
1 Frozen target, authenticated handoff, and no backflow	1
1.1 Dependency order	1
1.2 Exact import object	1
1.3 Theorem-use sources	2
1.4 No-backflow and target-change rules	2
2 Thirteen target fibres and the complete overlap graph	3
2.1 Antecedent-fibred comparison groupoids	3
2.2 The empty internal-gauge coordinate	5
2.3 Common bases and restrictions	5
2.4 Coordinate-factorization proof	6
3 Positive population of the thirteen-fibre joint object	7
3.1 The canonical predecessor tuple	7
3.2 Twenty edge witnesses	7
3.3 Source, domain, and carrier-sensitive checks	9
4 Witness-reflecting completion, complete certificates, probes, and lawful skin	10
4.1 The populated witness-reflecting incidence schema	10
4.2 Tuple-indexed forward-cosieve completion	10
4.3 The complete-certificate assembly object	11
4.4 Tensor, fibre, incidence, and complete-certificate probes	12
4.5 Certificate skin and thirteen fibre skins	13
5 Sorted candidate grammar and complete staged classification	14
5.1 Generated candidates and their sorts	14
5.2 Fibre status and tuple compatibility are different maps	15
5.3 Representative route classification	15
6 Thirteen independently tested fibre-finality theorems	17
6.1 Broad pre-finality domains	17
6.2 The thirteen response exhaustions	20
7 Compatibility descent, live residue, and the lawful orbit	23
7.1 Compatibility descends to fibre skin classes	23
7.2 Residue before canonical identification	23
7.3 Structural role signatures	24
7.4 Automorphisms of the actual complete certificate	25
8 Physical quotient, time and record standing, and WDW lineage	25
8.1 The physical object inside the joint certificate	25
8.2 Four temporal notions	26

8.3	Constraint-native realization of the WDW equation	27
9	Failure boundary and complete realization closure	27
9.1	Fail-closed domain	27
9.2	Certificate completeness and realization licensure	28
9.3	Endpoint payload and decision	29
9.4	Reportability matrix	29
10	Certified quantum-standing closure, occupied export, and downstream boundary	30
10.1	Certified representative interface and structural standing closure	30
10.2	Kernel-governed certified standing descent	31
10.3	The occupied WDW certificate and its comparison class	32
10.4	Exact non-export ledger	33
11	Dependency audit, gate matrix, and robustness analysis	33
11.1	Nineteen-node theorem chain	33
11.2	Gate matrix	34
11.3	Adversarial objections and exact resolutions	35
11.4	Publication verification	38
	References and source lineage	39
	Reference and source-lineage note	39

1. Frozen target, authenticated handoff, and no backflow

1.1 Dependency order

The proof inherits the one-way order

$$G_0 \longrightarrow G_\nabla \longrightarrow G_1 \longrightarrow G_2 \longrightarrow G_3 \longrightarrow G_4 \longrightarrow G_5 \longrightarrow G_6 \longrightarrow G_7 \longrightarrow G_8 \longrightarrow G_9.$$

Here G_0 is the nondegenerate construction kernel, G_∇ the accepted standing-preserving transport branch, and G_k the corresponding WDW endpoint. The arrows are theorem-use dependencies, not equivalences and not a chronological suggestion. No statement proved in this paper can make an earlier constraint algebra close, populate an earlier state, repair an operator domain, create a clock, or change a predecessor failure surface.

Definition 1.1 (Frozen core target). The object $\mathcal{D}_{\text{WDW}}^{\text{core}}$ consists of the following standing-bearing coordinates.

1. $\Sigma = \mathbb{H}^3/\Gamma$ is compact, connected, oriented, and boundaryless, with the accepted regular Sobolev/tame metric-scalar towers.
2. The classical point is the WDW-4A nonvacuum datum

$$z_\star = (h_\star, \pi_\star; 0, \epsilon_\star \sqrt{h_\star}), \quad \epsilon_\star > 0,$$

together with its structured source-attached Einstein–Klein–Gordon development $[\gamma_\star]_{\text{src}}$ on the licensed clock interval.

3. The canonical coordinate contains the WDW-1 metric-scalar phase carrier, symplectic form, Hamiltonian and momentum constraints, source-preserving split and reconstruction maps, and compact-core boundary ledger.
4. The quotient coordinate contains the WDW-2 proper spatial groupoid and regular quotient, and the WDW-3 full characteristic/refoliation groupoid with its common functional domain, singular guards, and failure boundary.
5. The quantum-candidate coordinate is the WDW-5 Branch-A configuration chart, separating test algebra, first and second Fréchet calculus, smoothing family, CCR seed, regulated constraint grammar, and common algebraic domain.
6. The quantum-survivor coordinate is the accepted WDW-6 counterterm-enriched zero-theta successor endpoint, complete constraint/anomaly closure, normalized nonzero joint solution, positive rank-one quotient, represented algebra $\mathbb{C}I$, and guarded WDW-7/8 interfaces.
7. The relational coordinate is the function-valued WDW-4A clock $\phi(\cdot) = \tau \mathbf{1}$, the WDW-7 quotient-native Cauchy doublet, constraint-induced connection, zero covariant line Hamiltonian, stationary physical ray, and nonconstant complete volume record.
8. The recovery coordinate is the WDW-8 source-attached single-rate germ, evaluated characteristic plaque, actual WDW-7 lineage, terminal zero exchange ledger, history-indexed strict cubes, five error coordinates, singleton lawful recovery residue, and exact failure/nonclaim package.
9. The local internal-gauge label set is $\mathcal{L}_{\text{int}} = \emptyset$. This is an empty typed factor of the real-scalar core, not Standard-Model content and not a zero operator substituted for an unconstructed gauge theory.

The list above fixes the endpoint being tested and identifies its imported canonical inhabitant; it is not the membership predicate for every comparison candidate. Section 2 separates each role into an antecedent contract α_i and a response type Ω_i . The displayed connection, development, operator, closure, quotient, clock and recovery values are then outputs to be recovered by the role theorems. They do not enter TESTABLE status or the broad pre-finality domains by being printed in the target.

The target is deliberately narrower than the union of every construction in the predecessor corpus. In particular, WDW-4B is a distinct tetrad/spin/gauge/ fermion enrichment. It does not enter Definition 1.1.

1.2 Exact import object

Let x_i^{pre} denote the accepted occupant assigned to the i th target role in Section 2. Define

$$I_{8 \rightarrow 9} := (\mathcal{D}_{\text{WDW}}^{\text{core}}, \{G_0, G_\nabla, G_1, \dots, G_8\}, \{\mathfrak{F}_i^{\text{pre}}, x_i^{\text{pre}}\}_{i=1}^{13}, \text{OverlapSeed}_9, \text{Fail}_{\leq 8}, \text{Nonclaim}_{\leq 8}, \text{Auth}_9 = \text{TRUE}). \quad (1)$$

The overlap seed records source and interface maps already defined upstream. It does not assert that all thirteen values commute simultaneously. Likewise, the handoff contains no WDW-9 completion colimit, joint probe, skin relation, candidate exhaustion, live residue, assignment space, or endpoint verdict.

The WDW-8 export retained in (1) is

$$G_8 := (\text{Scope}_8, \text{Imports}_8, \text{Pop}_8, \text{HJ}_8, \text{Cov}_8, \text{Mat}_8, \text{ObsRec}_8, \text{Back}_8, \text{Limit}_8, \text{Residue}_8, \text{Boundary}_8, \text{Cert}_8). \quad (2)$$

All twelve entries are imported. Keeping only the five displayed error values while dropping their source, continuation, topology, exchange, failure, or report coordinates would change the target.

Theorem 1.2 (W9.1 – Exact WDW-8 handoff and frozen target). *Equation (1) is the unique authorized WDW-9 entry object. It is nonempty, source-attached, branch-typed, and fixed at Definition 1.1. Every source, failure, and nonclaim coordinate of WDW-1–8 is retained. No WDW-9 conclusion is an antecedent of $I_{8 \rightarrow 9}$.*

Proof. The WDW-8 endpoint theorem proves

$$G_8 = \text{PASS}, \quad |\text{Real}_8^{\text{law}} / \sim_8| = 1, \quad \text{Authorize}(\text{WDW-9}) = \text{TRUE}$$

at precisely the target of Definition 1.1. Its proof cites positive family population, five typed error theorems, fixed-domain residue classification, and scope-safe continuation; it explicitly exports the lawful subtype, failure data, and nonclaims. The source-lineage entries in Table 1 identify the theorem objects used here. Since the right-hand side of (1) contains only those imported objects and their already defined interface maps, the definition neither assumes joint compatibility nor refers to a WDW-9 residue or orbit. This gives the claimed entry and acyclicity. $\square$

1.3 Theorem-use sources

Table 1: Source identity and theorem-use role.

ID	Source	Load-bearing use
KERNEL	Nondegenerate kernel	Adm/Standing/Reference/Irreversibility necessity
FIXED	Fixed-domain exhaustion	Complete same-target disposition grammar
SAMESCOPE	Same-scope operator exhaustion	No hidden operator or repair route
BIVALENCE	Bivalence theorem	Total PASS/FAIL status at each fixed gate
CLOCK	Physical clock quotient	Clock role, descent, and use-relative identity
MODULES	Modules A–D closure	Completion, probe, skin, residue, and orbit method
ARC	WDW ARC publication contract	Thirteen roles and WDW-9 endpoint
WDW1	Canonical projection	Connection, ADM carrier, source reconstruction, boundary
WDW2	Proper descent	Spatial groupoid, moment map, local quotient
WDW3	Constrained refoliation	HDA, characteristic equality, evaluated plaques
WDW4	Classical matter occupancy	Nonvacuum source, scalar clock, right inverse
WDW5	Candidate family	Configuration differential, grammar, common domain
WDW6	Physical realization	Renormalized closure, joint state, form, quotient, algebra
PHASEJET	Phase-jet sufficiency	First-jet recovery seed sufficiency
WDW7	Profile-clock exhaustion	Conditioning, connection, record, clock boundary
WDW8	Semiclassical recovery	Recovery, continuation, singleton residue, authorization

Source lineage establishes provenance, not mathematical truth. Every load-bearing use is restated and proved at the WDW-9 level where the predecessor theorem does not already supply the required conclusion.

1.4 No-backflow and target-change rules

Proposition 1.3 (Acyclic use). *No WDW-9 completion, probe, skin, residue, or orbit statement can authorize or repair a premise in $G_0, G_\nabla, G_1, \dots, G_8$.*

Proof. Every WDW-9 constructor has domain in the imported object (1) or in an object built from it by a source-preserving restriction, quotient, factorization, completion, or report map. None has codomain in the

truth-value coordinate of an upstream theorem. A failed upstream condition is carried by $\text{Fail}_{\leq 8}$ and makes the corresponding WDW-9 candidate ineligible. The bivalence interface prevents a third status in which a later object repairs the earlier failure while preserving the same lineage. Hence all arrows point downstream. $\square$

The following distinctions are therefore fixed throughout:

- a new connection, boundary generator, matter model, clock, ordering, regulator, counterterm, anomaly class, Hilbert topology, recovery carrier, or probe is an openly changed target;
- a proper spatial or refoliation representative change is potential skin only after every complete probe is preserved;
- a WDW-8 recovery element is a comparison object on $\{1\}_{\text{phys}} \sqcup (0, \delta_0]_{\text{rec}}$, not a second WDW-6 physical state;
- compact boundarylessness means that the core has no boundary generator; it does not imply that boundary time or charges lack standing on another target; and
- the empty Branch-A Gauss family neither imports WDW-4B nor proves a Standard-Model anomaly theorem.

Nonclaims fixed at entry. The paper does not prove Branch-B closure, a global clock, a preferred foliation, a nonzero full-Cauchy Hamiltonian, full-superspace or ultraviolet completion, topology change, singularity resolution, scattering, Poincaré or CPT recovery, cosmological branch selection, decoherence, an arrow of time, or phenomenology. It does not derive factor ordering, counterterms, G , Λ , masses, charges, mixings, or RG values from AASC alone.

2. Thirteen target fibres and the complete overlap graph

2.1 Antecedent-fibred comparison groupoids

For each role r_i , let $\mathfrak{M}_i$ be the generic interface-moduli groupoid of objects with the appropriate carrier type. Split the frozen role contract as

$$\theta_i = (\alpha_i, \Omega_i).$$

Here α_i contains only antecedent data: role and carrier type, source and branch, input or initial data, operator domain, boundary, declared parameters, operational burden, and the failure/report protocol. The symbol Ω_i is the *type* of the response coordinates to be tested. It does not contain the canonical response value $\omega_i^* := \hat{\Theta}_{9,i}(x_i^*)$. Define

$$\mathfrak{F}_i(\theta_i) := (\mathfrak{M}_i)_{/\alpha_i}^{\Omega_i},$$

the full groupoid of interface-eligible generated candidates lying over the same antecedent and carrying a response of the declared type. Its arrows are lawful role-preserving interface isomorphisms. Objects of this groupoid may have different response values and need not be connected to the intended occupant. Thus membership in $\mathfrak{F}_i(\theta_i)$ cannot imply equality under the later augmented probe.

This input/output separation is load-bearing. Antecedent fidelity is tested before comparison; the predecessor and overlap theorems must still prove which response classes survive. In particular, no singleton cardinality is encoded in θ_i , $\mathfrak{F}_i(\theta_i)$, or the definition of a candidate.

Definition 2.1 (Thirteen forced roles and canonical occupants). The forced role set is $R_9 = \{r_1, \dots, r_{13}\}$. The corresponding canonical occupants are:

ID	Fibre	Canonical occupant	Antecedent lock; response coordinates tested later
F1	Kernel/admissibility	$x_1^* = K_0$	Fixed nondegenerate comparison act and report protocol; Adm/Standing/Reference/Irreversibility response and no-repair status.
F2	Spacetime transport/connection	$x_2^* = C_{\nabla}^{\text{LC}}$	Standard local first-jet finite-linear metric-compatible torsion-free branch, regularity, recovery guards and no-selector condition; connection, derivative, curvature, holonomy and incident uses.

ID	Fibre	Canonical occupant	Antecedent lock; response coordinates tested later
F3	Covariant Einstein/source	$x_3^* = C_{\text{EKG}}$	Nonvacuum structured source ancestry, attachment certificate, Cauchy datum, orientation, interval and boundary; development, tensor witness and Ward/exchange response.
F4	Canonical carrier/boundary	$x_4^* = C_{\text{ADM}}$	Metric-scalar Cauchy carrier, source and compact-boundaryless contract; Legendre image, symplectic data, constraints and reconstruction.
F5	Spatial-diffeomorphism descent	$x_5^* = C_{\text{prop}}$	WDW-2 regular proper-action antecedent and charge/sector guards; zero locus, slice quotient, reduced form and descended source/observables.
F6	Internal-gauge/Gauss	$x_6^* = C_{\text{OG}}$	Branch-A real-scalar core with no WDW-4B carrier; internal-label family, Gauss action and incidence response.
F7	Refoliation/HDA	$x_7^* = C_{\text{HDA}}$	Fixed constraint map, source, clock patch, regular plaque and failure boundary; HDA, characteristic distribution, finite-route and continuation response.
F8	Configuration differential/representation	$x_8^* = C_{\text{conf}}$	Branch-A carrier, Sobolev chart, separating algebra, bosonic configuration polarization, smoothing family, CCR normalization, regulated grammar, proper quotient and domain; differentials, principal symbols, representation action and comparison maps.
F9	Ordering, regulator, domain	$x_9^* = C_{\text{ren}}$	Accepted WDW-5 ordering, heat-regulator grammar, finite-normalization profile, candidate syntax, source and regulator-free endpoint burden; local counterterm coefficients, successor graph, capwise operators, completion, common domain and endpoint response.
F10	Anomaly/closure	$x_{10}^* = C_{\text{anom}}^0$	Branch-A zero-theta constraint family and declared stronger-claim exclusions; closure and total-anomaly response.
F11	Physical inner product/observable	$x_{11}^* = C_{\text{phys}}$	Authenticated WDW-6 endpoint, normalized terminal seed and continuation interfaces; form, radical, quotient and represented algebra.
F12	Clock/conditional dynamics	$x_{12}^* = C_{\text{clk}}$	WDW-6 physical line, source history, clock-window and no-spectator burden; clock quotient, Cauchy doublet, connection, propagator, Hamiltonian and record use.
F13	Record, continuation, reconciliation	$x_{13}^* = C_{\text{rec}}$	Fixed WDW-8 handoff, source, single-rate scope and recovery domain; record, lineage, RPI, exchange, cubes, five errors, failure and nonclaim response.

Theorem 2.2 (W9.2 – Thirteen-fibre typed registry). *The thirteen antecedent-fibred groupoids $\mathfrak{F}_i(\theta_i)$ of Definition 2.1 are fixed before WDW-9 candidate inspection. Each has the displayed accepted occupant x_i^* , but no quotient cardinality is fixed by that fact. The role list is complete for the core ARC target and contains no duplicated physical role.*

Proof. The updated ARC target fixes the thirteen roles independently of the WDW-9 residue. WDW-1 supplies F2–F4; WDW-2 supplies F5; the Branch-A clauses of WDW-1–5 supply F6; WDW-3 supplies F7; WDW-5 supplies F8; WDW-5/6 supply F9; WDW-6 supplies F10–F11; WDW-4A/7 supply F12; and WDW-7/8 supply F13. F1 is the inherited governance type instantiated by every determinate use but kept as a separate unary role so it cannot be mistaken for a physical population theorem.

Completeness follows by taking the coordinate inventory of Definition 1.1. Every coordinate has exactly one primary role: governance; connection; covariant source; canonical carrier; proper spatial descent; internal

gauge; refoliation; configuration differential; ordering/domain; anomaly; physical quotient; clock/conditioning; or record/continuation/recovery. A purported fourteenth same-target role either is a coordinate inside one of these packages, duplicates an existing role, or introduces a new carrier, classifier, boundary, source, operator, clock, or use and hence changes the target. Existence of each displayed occupant is an imported positive theorem, not an inference from completeness of the list. $\square$

2.2 The empty internal-gauge coordinate

For the real-scalar core, put $\mathcal{L}_{\text{int}} = \emptyset$. Then

$$\bigcap_{\alpha \in \mathcal{L}_{\text{int}}} \ker \hat{G}_\alpha = \mathcal{H}_{\text{ambient}}$$

by the empty-intersection convention, and the internal-incidence monoid has its unique neutral element. This is a populated *typed absence*: the role records that no local internal gauge factor is present. Replacing it by a nonempty group, importing WDW-4B, or inserting a zero Gauss operator changes θ_6 and cannot be a second presentation of x_6^* .

2.3 Common bases and restrictions

For each generating edge $e = (i, j)$ let $\mathfrak{B}_e$ be the inherited common base type and let

$$q_{e,i} : \mathfrak{F}_i(\theta_i) \longrightarrow \mathfrak{B}_e, \quad q_{e,j} : \mathfrak{F}_j(\theta_j) \longrightarrow \mathfrak{B}_e$$

be the corresponding coordinate restrictions. Equality in $\mathfrak{B}_e$ is literal unless an upstream lawful quotient is part of the base type. No new quotient is introduced by this definition.

Table 3: Generating overlap bases and equality predicates.

Edge	Fibres	Common base	Equality predicate
E01	F2–F3	Covariant derivative/source	Same connection, derivative, curvature, source development, branch and regularity guard.
E02	F2–F13	Recovered connection	WDW-8 recovered derivative and curvature equal the F2 values on the exact recovery subtype.
E03	F3–F4	Covariant/canonical Cauchy data	Same metric, scalar, momenta, orientation, structured source class, attachment graph and Cauchy history.
E04	F3–F13	Comparator/source	Independent WDW-8 reconstruction and total source map equal the accepted F3 class in the declared topologies.
E05	F4–F5	Spatial moment map	F5's moment map and zero locus are precisely the F4 momentum constraint with the same sign, density and domain.
E06	F4–F6	Internal constraint slot	Both carry the empty Gauss label set and no Branch-B coordinate.
E07	F4–F7	Constraint generators	F7 acts through the exact F4 Hamiltonian and momentum densities on their common functional domain.
E08	F5–F7	Quotient and characteristic compatibility	Proper spatial and refoliation arrows preserve the same source, reconstruction, observable, boundary, failure and report coordinates.
E09	F5–F8	Quantum configuration descent	The Branch-A chart and test algebra descend under the accepted proper groupoid and no larger quotient.
E10	F6–F8	Empty internal representation	The Branch-A quantum grammar has no local Gauss generator or WDW-4B field.
E11	F7–F8	Classical principal symbol	The regulated Hamiltonian candidate has the accepted WDW-3/4A characteristic and density lineage.
E12	F8–F9	Operator grammar/-domain	Ordering, smoothing, counterterms and successor topology act on the exact F8 syntax and its common domain.
E13	F9–F10	Regulator/anomaly endpoint	Same counterterm successor carrier, graph domain, theta value, branch guard and regulator endpoint.

Edge	Fibres	Common base	Equality predicate
E14	F10–F11	Joint kernel/images	The normalized state annihilates the complete closed constraint system and the physical form is null on all its images.
E15	F11–F12	Physical conditioning	The WDW-7 line is the isometric image of the F11 quotient with its form and represented algebra preserved.
E16	F11–F13	Physical anchor/recovery	The exact physical anchor is retained; recovery elements remain on the disjoint comparison component and are not promoted to physical states.
E17	F12–F13	Clock/lineage/record	Same function-valued clock, actual WDW-7 lineage, connection, complete record and failure boundary.
E18	F7–F13	Characteristic continuation	The RPI seed is generated by WDW-4A spacetime flow through WDW-3 evaluated-plaque equivalence; the canonical formula is derived afterward.
E19	F3–F12	Source-attached clock history	Clock intersections and complete records are evaluated on the same licensed structured Einstein–Klein–Gordon history.
E20	F4–F13	Canonical recovery/-source response	WDW-8 Hamilton–Jacobi, tangent response and source corrections use the exact F4 canonical carrier and constraint map.

Define the pre-quotient joint *groupoid* as the full subgroupoid

$$\mathfrak{J}_9 = \left\{ (x_1, \dots, x_{13}) \in \prod_{i=1}^{13} \mathfrak{F}_i(\theta_i) : q_{e,i}(x_i) = q_{e,j}(x_j) \text{ for all } e \in E_9 \right\}. \quad (3)$$

Its arrows are tuples $(u_1, \dots, u_{13})$ of fibre arrows for which every restriction square in $\mathfrak{B}_e$ commutes and all source, domain, boundary, failure, and report locks are preserved. Hence the projections $\text{pr}_i : \mathfrak{J}_9 \rightarrow \mathfrak{F}_i(\theta_i)$ and the tuple-to-interface construction used below are actual functors, not set-level notation promoted after the fact.

2.4 Coordinate-factorization proof

To prove that Table 3 is complete, separate the complete target coordinates into four blocks:

$$\begin{aligned} \mathcal{U}_{\text{geo}} &= (\nabla, \text{Src}, \text{Cauchy}, \Phi_{\text{can}}, J_{\text{sp}}, \mathcal{L}_{\text{int}}, \text{HDA}, \partial\Sigma), \\ \mathcal{U}_{\text{q}} &= (\mathcal{U}_5, \mathcal{A}_5, d, d^2, \mathfrak{o}, K_\epsilon, \Phi_5, \mathcal{D}_{\text{gr}}, \text{Anom}, \theta), \\ \mathcal{U}_{\text{phys}} &= (F_A^{(0)}, B_A, \text{Rad } B_A, \mathcal{H}_{\text{phys}}, \mathbb{C}I, \phi = \tau \mathbf{1}, J_\tau, \nabla^C, H_{\text{rel}}^C, V^{\text{rec}}), \\ \mathcal{U}_{\text{rec}} &= (\mathcal{P}_8, \text{RPI}, \text{Plaque}, \text{Lineage}, \text{Exchange}, \text{Cube}, E_{\text{HJ}}, E_{\text{Ein}}, E_{\text{mat}}, E_{\text{obs}}, E_{\text{back}}). \end{aligned}$$

Every block also carries its source, boundary, failure and report subcoordinates. F1’s kernel profile is a unary response test on each nondegenerate block and therefore is not duplicated as pairwise incidence.

Lemma 2.3 (Shared-coordinate factorization). *Let u be a standing-bearing coordinate occurring in two or more fibres of Definition 2.1. Then equality of all appearances of u factors through an edge of Table 3 or through a composite of those edges. Source, boundary, failure, and report subcoordinates factor with their parent coordinate.*

Proof. For $\mathcal{U}_{\text{geo}}$, connection/source occurrences factor through E01–E04; canonical and spatial occurrences through E03, E05–E08; and clocked source or recovery occurrences through E04, E18–E20. The internal slot is connected to the canonical and quantum grammars by E06 and E10. Hence every geometric repetition is represented.

For $\mathcal{U}_{\text{q}}$, proper descent enters by E09, the classical principal symbol by E11, the operator syntax/domain by E12, the regulator and anomaly endpoint by E13, and the complete joint kernel by E14. Any repetition between nonadjacent quantum fibres is the composite of these consecutive restrictions.

For $\mathcal{U}_{\text{phys}}$, the physical quotient enters conditioning by E15 and recovery by E16; clock, lineage and record data enter recovery by E17; the structured history is synchronized by E19. Thus a repeated physical or clock coordinate factors through E15–E19.

For $\mathcal{U}_{\text{rec}}$, recovered connection, source, canonical, physical, clock and characteristic coordinates enter through E02, E04, E16–E18 and E20. The remaining RPI, exchange, cube and five-error data occur primarily in F13 and are unary target coordinates, not omitted pairwise incidences.

Finally, the predecessor definitions attach source ancestry, boundary status, first-failure status and reportability to the object being restricted. The restriction maps in Table 3 retain those fields. Therefore their equality follows through the same edge rather than through an unstated global identification. The four blocks exhaust Definition 1.1, so there is no further target-relevant shared coordinate. $\square$

Theorem 2.4 (W9.3 – Complete generating-overlap theorem). *The set $E_9 = \{E01, \dots, E20\}$ generates every target-relevant overlap among the thirteen fibres. A purported additional same-target overlap is either a composite of the displayed restrictions, presentation bookkeeping, an inadmissible hidden constructor, or an explicit target enrichment.*

Proof. Lemma 2.3 proves coverage of every shared standing-bearing coordinate. Conversely, each displayed edge is necessary: deleting it permits its common-base value to vary independently while leaving the remaining restrictions unchanged. For example, deletion of E03 permits a bare stress tensor to replace the structured source class; deletion of E12 permits an ordering or domain to detach from the candidate grammar; deletion of E16 permits comparison elements to be reclassified as physical states; and deletion of E17 permits a clock-synchronized but lineage-incompatible recovery. Such variations are detected by the frozen target and are not presentation bookkeeping. Thus the list is both sufficient and irredundant at the level of generating common bases. $\square$

3. Positive population of the thirteen-fibre joint object

3.1 The canonical predecessor tuple

Let

$$\Xi_9^* = (K_0, C_{\nabla}^{\text{LC}}, C_{\text{EKG}}, C_{\text{ADM}}, C_{\text{prop}}, C_{\text{ZG}}, C_{\text{HDA}}, C_{\text{conf}}, C_{\text{ren}}, C_{\text{anom}}^0, C_{\text{phys}}, C_{\text{clk}}, C_{\text{rec}}). \quad (4)$$

Every entry is an independently accepted object or the independently typed empty coordinate of the core. Equation (4) defines an element of the product of the thirteen fibres. It does not, by notation alone, establish membership in the equalizer conditions of (3).

3.2 Twenty edge witnesses

Table 4 records the literal predecessor results used for the joint check. “WDW-8 recovery” refers to the numbered W8 theorem burden in the final capstone; the detailed source and section locations are retained in the overlap audit ledger.

Table 4: Equality witnesses for Ξ_9^* .

Edge	Two restricted values	Equality witness
E01	∇^{LC} with its curvature and the derivative used by $[\gamma_\star]_{\text{src}}$	WDW-1 W1.1 fixes the unique minimal metric branch; WDW-4 W4.2 uses that connection in the scalar action, source, and Ward identity.
E02	F2 Levi–Civita profile and WDW-8 recovered connection/curvature	WDW-8 W8.9–W8.10 reconstruct the same source history and prove exact fixed-connection and curvature recovery on the stated subtype.
E03	Structured covariant Cauchy tuple and ADM split/reconstruction tuple	WDW-1 W1.5 gives a fibrewise local reconstructive equivalence preserving source class, attachment certificate, and witness identically; WDW-4 W4.2–W4.3 place $z_\star$ on that same fibre.
E04	F3 source/history and WDW-8 comparator/source response	WDW-8 W8.7–W8.9 reconstruct independently and prove equality with $[\gamma_\star]_{\text{src}}$; W8.13 proves exact history-dependent source attachment and Ward/exchange closure.
E05	F4 momentum constraint and F5 spatial moment map	WDW-2 Proposition 18.2 identifies the regular spatial moment-map zero locus with the WDW-1 momentum constraint, using the sign and density fixed by W1’s constraint identity.
E06	F4 internal slot and F6 Gauss label family	The Branch-A WDW-1/4 package contains no local internal gauge carrier; WDW-2 Definition 10.1 registers the occupied trivial factor. Both restrictions are $\emptyset$.

Edge	Two restricted values	Equality witness
E07	F4 Hamiltonian/momentum densities and F7 deformation generators	WDW-3 W3.3–W3.4 prove the exact off-shell algebra and full characteristic equality on the common functional algebra inherited from WDW-1; WDW-4 W4.5 occupies the scalar antecedent.
E08	F5 proper quotient and F7 characteristic/refoliation quotient	WDW-2 W2.7–W2.8 and WDW-3 W3.6–W3.8 prove source, observable, boundary and clock coherence on their registered intersection; WDW-4 W4.5 occupies that intersection.
E09	F5 proper action on the configuration carrier and F8 descended chart/test algebra	WDW-5 W5.3 restricts the exact accepted proper groupoid to the Branch-A configuration scale, proves the derivative loss, and proves local invariant descent without enlarging the quotient.
E10	F6 empty internal action and F8 Branch-A grammar	WDW-5's Branch-A linear/Hamiltonian candidate family consists of the spatial and Hamiltonian constraints; internal gauge and Lorentz generators belong only to Branch B. The two internal label sets are literally empty.
E11	F7 classical characteristic and F8 Hamiltonian principal symbol	WDW-5 W5.10 uses the exact WDW-1 normalization and complete scalar-clock classical terms; its regulator acts only on the quantum differential. WDW-8 W8.5–W8.7 recovers the inherited characteristic from the leading phase.
E12	F8 operator syntax/domain and F9 counterterm successor route	WDW-5 W5.10–W5.11 construct the complete regulated grammar and common algebraic domain; WDW-6 W6R.S12–S15 applies the finite-cap/counterterm successor construction to that same grammar and transports its graph domain.
E13	F9 successor endpoint and F10 anomaly endpoint	The WDW-6 renormalized closure uses the same counterterm-enriched carrier, successor graph topology, common regulator, ordering coordinates, and zero-theta branch in its complete constraint/anomaly calculation.
E14	F10 complete constraint family and F11 state/form	WDW-6's normalized $F_A^{(0)}$ lies in the joint annihilator of every complete renormalized constraint; Theorems W6R.20–W6R.21 make the terminal map and B_A null on every complete constraint image.
E15	F11 rank-one quotient and F12 conditioned line	WDW-7 W7.3–W7.5 proves that Cond_τ is an isometric unitary identification of the accepted WDW-6 quotient with the no-spectator Cauchy-doublet line and intertwines $\mathcal{C}I$.
E16	F11 physical anchor and F13 recovery carrier	WDW-8 fixes $\mathcal{P}_8 = \{1\}_{\text{phys}} \sqcup (0, \delta_0]_{\text{rec}}$, retains the exact WDW-6 anchor, and expressly proves no Hilbert, source, graph, or distributional convergence of recovery elements to the physical ray.
E17	F12 clock/lineage/record and F13 continuation package	WDW-8 W8.11–W8.14 uses the same profile clock and complete volume record and proves by guarded-intersection uniqueness that its evaluated line is the actual WDW-7 lineage.
E18	F7 evaluated characteristic and F13 RPI seed	WDW-4A supplies the smooth spacetime scalar flow; WDW-3 W3.5 supplies shifted-tower evaluated-plaque equivalence; WDW-8 W8.14 pushes the former through the latter before deriving the canonical lapse formula.
E19	F3 structured history and F12 clocked record	WDW-4 W4.6 constructs clock intersections on the licensed development of the same pointed source class; WDW-7 W7.6 evaluates the complete record on that occupied constrained orbit and preserves source ancestry.

Edge	Two restricted values	Equality witness
E20	F4 canonical carrier/constraint map and F13 HJ/source response	WDW-8 W8.5–W8.7 uses the WDW-5/6 principal symbol inherited from the exact F4 carrier; W8.13 constructs the tangent metric/source response on the same constraint map and local reconstruction chart.

3.3 Source, domain, and carrier-sensitive checks

Four groups of edges require more than equality of a headline output.

Structured source ancestry. On E03 the common value is not merely $T_{\mu\nu}$. It is

$$([\gamma_\star]_{\text{src}}, \hat{x}_\star, T_{x_\star}, \text{Att}_{x_\star}, W_{x_\star}; z_\star).$$

WDW-1’s split and reconstruction maps carry $(\hat{x}_\star, \text{Att}, W)$ identically. WDW-4 assigns the source from the covariant development rather than inferring it from a bare ADM tensor. WDW-8’s comparator and response constructions retain the same tuple. Thus E03, E04, E19, and E20 commute in the complete structured-source base.

Complete operator domain. E12 and E13 do not identify two differential formulas on a convenient core. Their common value contains the WDW-5 primitive-stable algebra, the complete constraint-label family, the counterterm-enriched successor carrier, graph topology, completion, and common composition domain. Killed exact unrenormalized lifts remain in the failure ledger and are not reintroduced by the equality.

Physical and recovery carriers. On E16 the common value is the *anchor coordinate*, not a state limit. The WDW-6 state is retained at 1_{phys} and the WDW-8 comparison elements live on $(0, \delta_0]_{\text{rec}}$. The coproduct tag is part of the base. Consequently the edge preserves rather than collapses their nonidentity.

Function-valued clock. E17 and E19 use $\phi(\cdot) = \tau \mathbf{1}$ together with its source history, orientation, transversality domain, quotient class, record use, and failure boundary. Replacing this by a spatial mean or scalar functional changes the base value and fails the edge.

Lemma 3.1 (All generating edges commute). *For every $e = (i, j) \in E_9$,*

$$q_{e,i}(x_i^\star) = q_{e,j}(x_j^\star) \quad \text{in } \mathfrak{B}_e.$$

Proof. The equality witnesses are given in Table 4. E03, E04, E19 and E20 have been checked in the complete source-attachment type; E12–E14 in the successor graph/domain and complete-constraint types; E15–E17 in the positive-quotient, disjoint-carrier and function-valued clock types; and E18 in the shifted-tower evaluated-plaque type. The remaining edges are literal equalities or equalities in the previously accepted proper/refoliation quotient. No row uses a WDW-9 probe, skin, residue, or assignment conclusion. $\square$

Theorem 3.2 (W9.4 – Positive joint-section theorem). *The tuple (4) is a member of (3). Hence*

$$\Xi_9^\star \in \mathfrak{J}_9, \quad \mathfrak{J}_9 \neq \emptyset.$$

The witness is nondegenerate.

Proof. Theorem 2.2 places every entry in its target fibre. Theorem 2.4 proves that E01–E20 are the complete generating equalizer conditions, and Lemma 3.1 proves all of them for $\Xi_9^\star$. This is exactly membership in (3).

Nondegeneracy is witnessed inside the same tuple: F3 contains the nonvacuum source history with $\epsilon_\star > 0$; F11 contains the normalized nonzero joint solution and nonzero positive quotient; F12 contains a nonconstant complete volume record; and F13 contains populated non-anchor recovery elements and a nonidentity upstream continuation category. These witnesses are retained by the overlap maps, so the joint object is not an empty or all-neutral product. $\square$

Corollary 3.3 (Authorization of the completion layer). *The source/interface schema induced by $\Xi_9^\star$ is populated. The completion, probe, and skin constructions of Section 4 may therefore be formed without assuming uniqueness or a final residue.*

Remark 3.4 (Failure discipline). Had any row of Table 4 failed, the first failed edge in the fixed order would have been a joint obstruction. Neither a colimit nor a later skin quotient would be permitted to identify the two values. The positive result above is therefore a gluing theorem, not a definition of the desired tuple as compatible.

4. Witness-reflecting completion, complete certificates, probes, and lawful skin

4.1 The populated witness-reflecting incidence schema

Let $\widehat{S}_9$ be the small category generated by:

1. the source, target, boundary, failure, and report sorts of each of the thirteen fibre presentations;
2. every primary predecessor interface internal to those presentations;
3. for each $e \in E_9$, a distinct base sort b_e , the two restriction legs into b_e , and the inherited maps out of that base;
4. the identities, inherited composites, and inherited equalities already proved in the accepted source categories.

No residue, skin relation, assignment, realization conclusion, or equality between unequal base values is a generator of $\widehat{S}_9$.

The witness object $\widehat{W}_9 \in \mathbf{Set}^{\widehat{S}_9}$ retains the actual value types, rather than merely a status label. In particular,

$$\widehat{W}_9(b_e) = \mathfrak{B}_e, \quad \widehat{w}_{e,i}(x_i) = q_{e,i}(x_i), \quad \widehat{w}_{e,j}(x_j) = q_{e,j}(x_j). \quad (5)$$

At all other sorts $\widehat{W}_9$ records the registered source identifier, target parameter, domain, boundary value, first-failure value, and report value in its actual ambient set. The canonical tuple selects a section $\widehat{w}_9^*$ of these values. A witness-locked interface is an object of

$$\widehat{\text{Int}}_9 := \mathbf{Set}^{\widehat{S}_9} / \widehat{W}_9. \quad (6)$$

Theorem 3.2 ensures that the two restriction legs over every b_e carry the same canonical value. Thus the actual forty legs define a diagram in (6); the slice is not being used to manufacture their agreement.

4.2 Tuple-indexed forward-cosieve completion

Fix $\mathbf{x} = (x_1, \dots, x_{13}) \in \mathfrak{J}_9$. For a primary interface occurrence $a : s \rightarrow t$, let

$$j_a : \widehat{S}_{9,a}^{\text{pri}} \hookrightarrow \widehat{S}_{9,a}^{\text{fwd}}$$

be the inclusion of its primary support into its forward cosieve. Let $D_9(\mathbf{x})$ denote the normalized registered-interface datum consisting of every primary interface coordinate, every $q_{e,i}(x_i)$, and the source, domain, boundary, failure and report locks. These are factors of the augmented fibre probes; no raw fibre-object name is a coordinate of D_9 . If $X_a(D_9(\mathbf{x})) \rightarrow \widehat{W}_9$ is the corresponding accepted datum, define

$$\overline{X}_a(\mathbf{x}) := \text{Lan}_{j_a} X_a(D_9(\mathbf{x})) \quad \text{in } \widehat{\text{Int}}_9. \quad (7)$$

Pointwise, for $u \in \widehat{S}_{9,a}^{\text{fwd}}$,

$$\overline{X}_a(\mathbf{x})(u) \cong \text{colim}_{(v, j_a v \rightarrow u)} X_a(\mathbf{x})(v). \quad (8)$$

A completed term is therefore a declared primary value followed by a directed incidence path. Paths are identified only by an inherited category law or a proved overlap equality.

Let $\mathcal{G}_9(\mathbf{x})$ consist of all completed primary occurrences, all completed overlap spans, and their induced maps. Its tensor/interface ledger is

$$L_9(\mathbf{x}) := \text{colim}_{a \in \mathcal{G}_9(\mathbf{x})} \overline{X}_a(\mathbf{x}) \quad \text{in } \widehat{\text{Int}}_9, \quad L_9^0 := L_9(\Xi_9^*). \quad (9)$$

The superscript in L_9^0 denotes the canonical ledger, not a complete thirteen-coordinate certificate.

Lemma 4.1 (Existence, pointwise form, and coverage). *The Kan extensions and colimits in (7)–(9) exist. At each sort s , $L_9(\mathbf{x})(s)$ is the disjoint union of the completed generator values modulo the least equivalence relation generated by the arrows of $\mathcal{G}_9(\mathbf{x})$. Every lawful completed interface term has a representative, and every representative comes from one registered primary occurrence and a forward path.*

Proof. The functor category $\mathbf{Set}^{\widehat{S}_9}$ is cocomplete, and its slice over $\widehat{W}_9$ is cocomplete with colimits created by the forgetful functor. Left Kan extensions along small functors exist and are computed by the pointwise formula (8). Set-valued colimits have the stated quotient presentation. The indexing set consists of interface occurrences, not interface names, so each primary occurrence enters once. The only nonidentity generating arrows are inherited interface arrows, forward incidence arrows, and E01–E20. This proves existence, coverage, and the least-relation assertion. $\square$

Proposition 4.2 (Value-reflecting no-repair theorem). *Suppose $e = (i, j)$ and $q_{e,i}(x_i) \neq q_{e,j}(x_j)$ in $\mathfrak{B}_e$. No cocone in $\widehat{\mathbf{Int}}_9$ can identify the corresponding two b_e -terms. Consequently such a tuple cannot enter $\mathfrak{J}_9$, and no lawful Kan extension or tensor colimit can repair its failed overlap.*

Proof. Let u and v be the two b_e -terms. By (5), their structure-map values are $q_{e,i}(x_i)$ and $q_{e,j}(x_j)$ as actual elements of $\widehat{W}_9(b_e) = \mathfrak{B}_e$. If a slice cocone sent u and v to the same element z , commutativity over $\widehat{W}_9$ would give

$$q_{e,i}(x_i) = \widehat{w}(z) = q_{e,j}(x_j),$$

a contradiction. The argument depends on the value-reflecting base sort; source locking by a singleton status object alone would not suffice. $\square$

Theorem 4.3 (W9.5 – Witness-reflecting completion and tensor colimit). *For every $\mathbf{x} \in \mathfrak{J}_9$, $L_9(\mathbf{x})$ is the populated witness-reflecting completion of its thirteen primary interface families and twenty overlaps. Every compatible cocone admits a unique mediator. The canonical ledger L_9^0 is populated by Ξ_9^* . Completion adds no operator, source, clock, classifier, physical occupant, or identification beyond the declared diagram, and it cannot erase an unequal common-base value.*

Proof. Lemma 4.1 supplies the colimit and universal mediator. The source, target, domain, boundary, failure, report, and twenty base-value coordinates are preserved because all maps are morphisms over $\widehat{W}_9$. Proposition 4.2 gives the standing-bearing separation. Population of L_9^0 follows from the actual interface diagram and cocone of Ξ_9^* . $\square$

4.3 The complete-certificate assembly object

The tensor ledger does not contain a thirteen-tuple. To assemble both levels, let $\mathfrak{B}_9^{\text{asm}}$ be the groupoid of registered interface diagrams together with their twenty base values and their source, domain, boundary, failure, and report locks. For $b \in \mathfrak{B}_9^{\text{asm}}$, let $\mathbf{LedPres}_9(b)$ be the groupoid of colimit presentations of the completed diagram encoded by b , including the universal cocone, its map to $\widehat{W}_9$, and the registered forward-probe cocone into P_9^{ten} . Its arrows are presentation isomorphisms commuting with all three pieces of data. Set

$$\mathfrak{L}_9 := \int_{b \in \mathfrak{B}_9^{\text{asm}}} \mathbf{LedPres}_9(b).$$

The Grothendieck projection and the tuple-to-diagram construction give two functors:

$$\beta_9 : \mathfrak{J}_9 \longrightarrow \mathfrak{B}_9^{\text{asm}}, \quad \mathbf{x} \longmapsto \mathcal{G}_9(\mathbf{x}), \quad (10)$$

$$\lambda_9 : \mathfrak{L}_9 \longrightarrow \mathfrak{B}_9^{\text{asm}}. \quad (11)$$

Thus an object of $\mathfrak{L}_9$ is a completed ledger presentation (b, L, κ) , and λ_9 returns exactly b . Choose the standard split Grothendieck presentation, so equality over the base is literal; equivalently, one may use the iso-comma pullback, which is groupoid-equivalent. Define the complete certificate presentation groupoid by the strict pullback

$$\mathfrak{C}_9^0 := \mathfrak{J}_9 \times_{\mathfrak{B}_9^{\text{asm}}} \mathfrak{L}_9. \quad (12)$$

Thus an object is

$$c = (\mathbf{x}, L, \kappa; \alpha),$$

where $\mathbf{x} \in \mathfrak{J}_9$, (L, κ) presents $L_9(\mathbf{x})$, and α is the displayed equality of assembly data. The pullback has genuine projections

$$\pi_{\mathfrak{J}} : \mathfrak{C}_9^0 \rightarrow \mathfrak{J}_9, \quad \pi_i : \mathfrak{C}_9^0 \rightarrow \mathfrak{F}_i(\theta_i), \quad \pi_L : \mathfrak{C}_9^0 \rightarrow \mathfrak{L}_9. \quad (13)$$

with $\pi_i = \text{pr}_i \pi_{\mathfrak{J}}$. No map $\mathfrak{F}_i(\theta_i) \rightarrow L_9^0$ is asserted or required.

The positive joint section and its tensor ledger give the canonical certificate

$$c_9^* = (\Xi_9^*, L_9^0, \kappa^*; \text{id}) \in \mathfrak{C}_9^0. \quad (14)$$

This is a population theorem prior to any skin or uniqueness conclusion.

4.4 Tensor, fibre, incidence, and complete-certificate probes

For every primary occurrence a , let P_a^{fwd} be its already registered forward probe object and write the accepted probe as a natural map

$$\vartheta_a : X_a \longrightarrow j_a^* P_a^{\text{fwd}}.$$

The Kan-extension adjunction gives the uniquely corresponding map

$$\bar{\vartheta}_a : \text{Lan}_{j_a} X_a \longrightarrow P_a^{\text{fwd}}.$$

The registered comparison maps $P_a^{\text{fwd}} \rightarrow P_9^{\text{ten}}$ make these maps a cocone on $\mathcal{G}_9(\mathbf{x})$; compatibility on the completed spans is exactly E01–E20 and the inherited naturality laws. The colimit property therefore supplies, for each ledger presentation $\ell = (b, L, \kappa)$, one mediator

$$\vartheta_\ell : L \longrightarrow P_9^{\text{ten}}. \quad (15)$$

This arrow is not itself a functor with domain $\mathfrak{L}_9$. To keep the two categorical levels distinct, define the mediator groupoid

$$\text{Med}_9 := \{(\ell, \vartheta_\ell) : \ell \in \mathfrak{L}_9, \vartheta_\ell : L_\ell \rightarrow P_9^{\text{ten}} \text{ is the induced mediator}\}. \quad (16)$$

A morphism $(\ell, \vartheta_\ell) \rightarrow (\ell', \vartheta_{\ell'})$ is a source-locked ledger-presentation isomorphism $u : L_\ell \rightarrow L_{\ell'}$ satisfying $\vartheta_{\ell'} u = \vartheta_\ell$. Let

$$\Pi_9^{\text{ten}} := \pi_0(\text{Med}_9)$$

and regard it as a discrete groupoid. Colimit functoriality sends every arrow $u : \ell \rightarrow \ell'$ in $\mathfrak{L}_9$ to the displayed commuting mediator isomorphism; uniqueness of the two colimit mediators proves the required equality. Hence

$$\Theta_9^{\text{ten}} : \mathfrak{L}_9 \longrightarrow \text{Disc}(\Pi_9^{\text{ten}}), \quad \ell \longmapsto [(\ell, \vartheta_\ell)] \quad (17)$$

is a genuine functor. It reads the completed ledger and mediator class only; it is not a product of thirteen fibre probes.

For each i , let P_i be the complete target-coordinate product named by Fi in Definition 2.1, and define

$$\Theta_i : \mathfrak{F}_i(\theta_i) \longrightarrow P_i$$

by simultaneous coordinate evaluation. Augment it by every incident restriction and local lock:

$$\widehat{\Theta}_{9,i}(x_i) = (\Theta_i(x_i), \{q_{e,i}(x_i)\}_{e \ni i}, \text{Source}_i, \text{Domain}_i, \text{Boundary}_i, \text{Failure}_i, \text{Report}_i) \in \widehat{P}_i. \quad (18)$$

For $e = (i, j)$ retain the pair-valued incidence probe

$$\Theta_e^{\text{inc}} : \mathfrak{F}_i(\theta_i) \times \mathfrak{F}_j(\theta_j) \longrightarrow P_e^{\text{inc}}, \quad \Theta_e^{\text{inc}}(x_i, x_j) = (q_{e,i}(x_i), q_{e,j}(x_j), \mathbf{1}[q_{e,i}(x_i) = q_{e,j}(x_j)], \text{FirstFail}_e). \quad (19)$$

Its domain is the indicated fibre product, not a single ledger term.

Now the projections (13) type the complete probe:

$$\begin{aligned} \Theta_9^{\text{cert}} : \mathfrak{C}_9^0 &\longrightarrow P_9^{\text{cert}}, \\ \Theta_9^{\text{cert}}(c) &= \left(\Theta_9^{\text{ten}}(\pi_L c), \{\widehat{\Theta}_{9,i}(\pi_i c)\}_{i=1}^{13}, \{\Theta_e^{\text{inc}}(\pi_i c, \pi_j c)\}_{e=(i,j)}, \text{AssemblyLock}(c) \right). \end{aligned} \quad (20)$$

The codomain is the displayed product of the tensor, augmented-fibre, incidence, and assembly-lock codomains, fixed before a skin or residue is formed.

Lemma 4.4 (Typed separation and projection). *If two complete certificates differ in any standing-bearing tensor, fibre, incidence, source, domain, boundary, failure, or report coordinate, their Θ_9^{cert} values differ. Equality of complete-probe values implies equality of every displayed probe factor and therefore projects to equality under each $\widehat{\Theta}_{9,i}$. No converse asserting the existence of a presentation arrow is made.*

Proof. Each listed coordinate is literally a factor of (20); every incident base value is also a factor of (18). The tensor factor is the mediator isomorphism class in (17); two nonisomorphic source-locked mediator diagrams therefore have different values. This proves separation. Equality of the complete tuple of factors gives the projection assertion directly. A kernel pair may relate two probe-indistinguishable objects even when no arrow between them was previously registered; none of the later arguments requires such an arrow. $\square$

Theorem 4.5 (W9.6 – Complete-certificate and joint-probe theorem). *The groupoid $\mathfrak{C}_9^0$ is positively populated by c_9^* . The tensor probe Θ_9^{ten} and complete probe Θ_9^{cert} exist with the domains displayed in (17) and (20); both are independent of a completed path representative. The complete probe detects every standing-bearing fibre or incidence difference at the frozen target.*

Proof. Population is (14). The Kan-extension adjunction and W9.5 colimit mediator construct (15); colimit functoriality and uniqueness then construct the functor in (17). The pullback projections construct every remaining coordinate of (20), so no missing projection from a ledger term is presumed. Passage to $\pi_0(\mathbf{Med}_9)$ makes the tensor coordinate independent of the ledger presentation, and Lemma 4.4 proves completeness. $\square$

4.5 Certificate skin and thirteen fibre skins

Define relations only on the objects on which their probes are defined:

$$S_9^{\text{cert}} := \text{KerPair}(\Theta_9^{\text{cert}} \Rightarrow \mathfrak{C}_9^0, \quad (21)$$

$$S_{9,i} := \text{KerPair}(\hat{\Theta}_{9,i} \Rightarrow \mathfrak{F}_i(\theta_i), \quad 1 \leq i \leq 13. \quad (22)$$

Lemma 4.6 (Congruence and projection). *S_9^{cert} and every $S_{9,i}$ are internal equivalence relations and congruences for their declared operations. Moreover,*

$$c S_9^{\text{cert}} c' \implies \pi_i(c) S_{9,i} \pi_i(c') \quad (1 \leq i \leq 13).$$

Proof. Kernel pairs are internal equivalence relations. Naturality of every declared operation with respect to the corresponding probe makes them congruences. The projection implication holds because $\hat{\Theta}_{9,i} \pi_i$ is a factor of Θ_9^{cert} . This proof uses only equality of probe values; it neither assumes nor constructs a groupoid arrow between every related pair. $\square$

Theorem 4.7 (W9.7 – Maximal lawful certificate and fibre skins). *S_9^{cert} is the unique maximal same-target standing-neutral congruence on complete-certificate presentations. For each i , $S_{9,i}$ is the unique maximal same-target standing-neutral response congruence on $\mathfrak{F}_i(\theta_i)$. Neither kind of skin identifies unequal incidence, source, physical, clock, recovery, failure, or report values.*

Proof. Let R be a standing-neutral congruence on the appropriate domain. Standing-neutrality says that the relevant complete probe coequalizes the two projections of R . The kernel-pair pullback property gives a unique map from R into the corresponding relation in (21) or (22); hence R is contained in it. Lemma 4.6 proves that each displayed kernel pair is itself lawful. Thus it is maximal and unique. The final statement follows because every prohibited coordinate is an explicit probe factor. $\square$

Corollary 4.8 (No physical erasure). *The quotient $\mathfrak{C}_9^0/S_9^{\text{cert}}$ preserves the structured source class, complete graph domain, zero-theta anomaly package, WDW-6 physical quotient, function-valued clock, nonconstant record, disjoint recovery carrier, exchange cube, and every failure/nonclaim coordinate. The quotient $\mathfrak{F}_i(\theta_i)/S_{9,i}$ preserves all standing-bearing F_i and incident values.*

Proof. Every listed coordinate is a factor of the corresponding probe. Quotienting by its kernel pair cannot identify unequal values of any factor. $\square$

Proposition 4.9 (Probe-kernel descent of interface completion). *Let $\mathbf{x}, \mathbf{y} \in \mathfrak{J}_9$ satisfy $x_i S_{9,i} y_i$ for every i . Then for universal ledger presentations ℓ_x and ℓ_y of $L_9(\mathbf{x})$ and $L_9(\mathbf{y})$,*

$$D_9(\mathbf{x}) = D_9(\mathbf{y}), \quad \Theta_9^{\text{ten}}(\ell_x) = \Theta_9^{\text{ten}}(\ell_y).$$

For any complete-certificate presentations c_x, c_y over these tuples with their universal ledger presentations, all remaining complete-probe factors agree and hence $c_x S_9^{\text{cert}} c_y$.

Proof. Each primary-interface coordinate, incident restriction and lock in D_9 is a factor of some $\hat{\Theta}_{9,i}$. Fibre-skin equality therefore gives literal equality of the two normalized interface diagrams. Their Kan extensions are the same diagram, and uniqueness of colimits supplies the unique source-locked isomorphism between any two presentations, commuting with the universal and registered probe cocones. Thus the mediator diagrams define the same element of Π_9^{ten} . The fibre factors agree by hypothesis, the incidence factors agree because every $q_{e,i}$ agrees, and the assembly locks belong to D_9 . Equation (20) now gives equality of the complete probes. No arrow between x_i and y_i is used. $\square$

5. Sorted candidate grammar and complete staged classification

5.1 Generated candidates and their sorts

The fixed candidate universe is the disjoint sorted class

$$\text{Cand}_9^{\text{fix}} = \text{Cand}_9^{\text{cert}} \sqcup \bigsqcup_{i=1}^{13} \text{Cand}_{9,i} \sqcup \text{Fail}_9 \sqcup \text{Reset}_9. \quad (23)$$

Each $\text{Cand}_{9,i}$ is typed over the generic role groupoid $\mathfrak{M}_i$; it is not assumed to lie in $\mathfrak{F}_i(\theta_i)$. Only a candidate that passes eligibility and *antecedent* fidelity enters that comparison fibre; its response value remains unrestricted. Likewise, $\text{Cand}_9^{\text{cert}}$ contains tuple/assembly routes before compatibility; only compatible all-testable tuples assemble into $\mathcal{C}_9^0$. It is the least such class containing $I_{8 \rightarrow 9}$ and closed under:

- C1.** extraction of a registered coordinate into its declared fibre sort;
- C2.** lawful redescription by an accepted presentation arrow in that sort;
- C3.** interface restriction with synchronized source, domain, boundary, failure, and report data;
- C4.** a registered quotient, universal factorization, Kan extension, tensor colimit map, or complete-certificate assembly map;
- C5.** incidence evaluation on E01–E20;
- C6.** formation of an explicit failure object or report object; and
- C7.** an explicit reset into a newly declared target certificate.

The class includes failed and changed-target routes. It does not set an error to zero, designate an intended occupant as live, or prescribe a residue cardinality.

An expression requiring a new operator, tensor, source, clock, selector, label, topology, probe, continuation rule, physical identification, or certificate projection is not silently a C1–C6 term. It is either a hidden load or an explicit C7 reset.

Definition 5.1 (Sorted normal forms). Every generated route is assigned one of:

- N1.** a canonical fibre coordinate in one $\text{Cand}_{9,i}$;
- N2.** an accepted same-fibre presentation of N1;
- N3.** a synchronized restriction of N1 or N2;
- N4.** a registered ledger or certificate universal image;
- N5.** a fibre, tensor, incidence, or complete-certificate evaluation;
- N6.** an explicit failure/report package with its first failed gate; or
- N7.** a changed-target package with the reset coordinates exposed.

Each normal form retains its sort from (23).

Lemma 5.2 (Structural normalization). *Every element of $\text{Cand}_9^{\text{fix}}$ has a sorted normal form. Two antecedent-faithful testable forms in $\mathfrak{F}_i(\theta_i)$ are related by $S_{9,i}$ exactly when their augmented response probes agree; this statement does not assert a presentation arrow between them. Failed, hidden-load, and reset forms are not quotiented by that relation. Two assembled complete-certificate forms are compared by S_9^{cert} . No certificate skin is applied directly to a fibre occupant.*

Proof. Induct on constructor depth. C1–C7 give N1–N7 respectively, with C4 and C5 retaining whether their result is a fibre, ledger, incidence, or certificate object. Normalize the inputs of a composite and apply its outer constructor. Functoriality and universal-factorization uniqueness remove path and parenthesization duplicates. W9.7 compares the remaining probe-invisible multiplicity, after eligibility and antecedent fidelity have placed the object in the probe domain, by the kernel pair matching that domain. The disjoint union (23) prevents a relation from being silently transported to another domain. $\square$

Theorem 5.3 (W9.8 – Candidate-grammar completeness). *Definition 5.1 exhausts every construction route available at the fixed WDW-9 target. A purported eighth same-target route is a duplicate normal form, an inadmissible hidden load, or an explicit target change.*

Proof. Lemma 5.2 proves completeness for the generated grammar. Fixed-domain exhaustion reduces an admissibility-bearing act to boundary typing, evaluated use, lawful quotient class, and continuation. Same-scope operator exhaustion separately covers carrier, grammar, domain, and operator change. C1–C6 contain the registered forms of all those acts. An extra constructor that changes no probe factor is presentation; one that changes a factor while retaining the old target label is a hidden load; a declared change is C7. Bivalence excludes an intermediate case. $\square$

5.2 Fibre status and tuple compatibility are different maps

For $x \in \text{Cand}_{9,i}$ define independently:

- $\text{Elig}_i(x)$: its carrier, interface direction, source/target types, and operational burden are appropriate for role r_i ;
- $\text{Fid}_i^{\text{ant}}(x; \theta_i)$: x lies over the frozen antecedent α_i and returns a response of type Ω_i ; this predicate does *not* require $\hat{\Theta}_{9,i}(x) = \hat{\Theta}_{9,i}(x_i^*)$;
- $\text{Form}_i(x)$: x is produced by the registered grammar with the declared source, domain, boundary, failure, and report interfaces; and
- $\text{FirstFormFail}_i(x)$: the least failed formation gate in the fixed order of imported, eligibility, antecedent-fidelity, source, domain, boundary, and typing tests.

Eligibility and antecedent fidelity are structural rather than equality to x_i^* . For example, a putative second local first-jet metric-compatible torsion-free connection satisfying the same no-selector burden, or a nonclosed zero-theta constraint response, can be testable at this stage. The response theorems in Section 6, not this definition, decide whether such candidates survive.

Definition 5.4 (Exclusive raw-fibre disposition). For $x \in \text{Cand}_{9,i}$, define $\text{Disp}_{9,i}(x)$ by exactly one of the following clauses:

1. TESTABLE, when x is eligible, antecedent-faithful, is formed by the registered grammar, and uses no hidden constructor; no response value is fixed by this status;
2. TARGET-RESET, when C7 exposes a changed target;
3. HIDDEN-LOAD, when an unregistered constructor is used under the unchanged target label; or
4. FAIL(g_k), where $g_k = \text{FirstFormFail}_i(x)$.

For a thirteen-tuple $\mathbf{x} = (x_1, \dots, x_{13})$ of testable fibre candidates, define

$$\text{Compat}(\mathbf{x}) = 1 \iff q_{e,i}(x_i) = q_{e,j}(x_j) \text{ for every } e = (i, j),$$

together with equality of the source, domain, boundary, failure, and report locks. Its certificate disposition is

$$\text{Disp}_9^{\text{cert}}(\mathbf{x}) \in \{\text{COMPATIBLE}, \text{FAIL}(E_k), \text{FAIL}(\text{lock } k)\}, \quad (24)$$

using the first failed edge or lock.

Lemma 5.5 (Totality, exclusivity, and first failure). *Every fibre candidate has exactly one disposition in Definition 5.4. Every all-testable tuple has exactly one certificate disposition in (24). In particular, $S_{9,i}$ and S_9^{cert} are equivalence relations, not raw-candidate dispositions; reflexivity therefore creates no status overlap.*

Proof. Normalize by W9.8. N7 gives an exposed reset. An undeclared constructor under the old target gives a hidden load. Otherwise apply the finite ordered local tests. Bivalence makes every test pass or fail, and the fixed order supplies the least formation failure. Passing the formation tests gives TESTABLE; none of those tests asks whether the response equals the canonical response. These cases are definitionally disjoint. For an all-testable tuple, apply the twenty edge tests followed by the lock tests in their fixed order; either the first one fails or all pass. Presentation equivalence is assessed only after raw status and never competes with it. $\square$

5.3 Representative route classification

Table 5: Sorted same-scope and changed-scope routes.

Route	Sort	Scope status	Quotient relation or later decisive test
Canonical predecessor tuple	certificate	COMPATIBLE	W9.4 and the canonical certificate (14).
Canonical fibre occupant	F_i	TESTABLE	Positively occupies its local target and one compatible certificate.
Proper spatial representative	F5	TESTABLE	$S_{9,5}$ -equivalent by the accepted WDW-2 presentation arrow.
Guarded refoliation representative	F7	TESTABLE	$S_{9,7}$ -equivalent with source, clock, record, and boundary fixed.
Recovery-rate reparameterization	F13	TESTABLE	$S_{9,13}$ -equivalent; all error and failure coordinates agree.
Constant phase-frame change	F13	TESTABLE	$S_{9,13}$ -equivalent; no phase derivative or incidence changes.
WDW-6 fixed-profile scale address	F11	TESTABLE	$S_{9,11}$ -equivalent only after complete state/-form/quotient matching.
Different structured source ancestry	reset	TARGET-RESET	F3/F4/F12/F13 and E03/E04/E19/E20 change.
Equal bare tensor, different attachment graph	F3	FAIL(E03)	Structured source base values are unequal.
Nonempty internal gauge family	reset	TARGET-RESET	F6 and E06/E10 change; this is an enrichment.
Declared different ordering/counterterm	reset	TARGET-RESET	F9 and E12/E13 change.
Old exact unrenormalized lift	F9	TESTABLE	Eliminated only by the W9.10 F9 regulator/obstruction test.
Same-label nonclosed anomaly	F10	TESTABLE	Eliminated only by the W9.10 F10 closure test.
Declared different anomaly target	reset	TARGET-RESET	Branch, theta, or closure target changes.
Additional physical state or form	reset	TARGET-RESET	F11 rank-one target and augmented probe change.
WKB recovery element offered as physical state	F11	TESTABLE	Eliminated only by E16 and the W9.10 F11 component-separation test.
Point-only conditional trace	F12	TESTABLE	Eliminated only by the W9.10 F12 nonlossy-trace test.
Scalar mean offered as clock	F12	TESTABLE	Eliminated only by the W9.10 F12 clock-exhaustion test.
External coordinate-time selector	hidden load	HIDDEN-LOAD	No registered F12 constructor supplies an independent selector.
Different record or lineage	reset	TARGET-RESET	F12/F13 and E17/E19 change.
Wrong-source exchange cell	F13	FAIL(F13-CUBE)	It is not history-indexed over the accepted source.
Undeclared tensor or selector	hidden load	HIDDEN-LOAD	No C1–C6 constructor and no exposed reset.
Boundary/asymptotic generator	reset	TARGET-RESET	The compact empty boundary ledger changes.
Branch-B/Standard-Model package	reset	TARGET-RESET	Carriers, Gauss family, domains, matter, and anomaly target change.
Global clock or singular continuation	reset	TARGET-RESET	Continuation domain and failure boundary change.

Theorem 5.6 (W9.9 – Eligibility, fidelity, compatibility, and failure completeness). *Every generated fibre candidate*

receives exactly one raw disposition, every all-testable tuple is classified as compatible or by its first failed edge or lock, and probe-indistinguishable same-target forms lie in the correctly typed relation $S_{9,i}$ or S_9^{cert} . No response survivor and no fibre cardinality is decided here. The words “live” and “skin” do not denote competing candidate statuses: live classes are formed only in Section 7, while skin denotes a binary relation.

Proof. W9.8 supplies the exhaustive sorted grammar. Lemma 5.5 proves status totality, exclusivity, and first failure. W9.7 supplies exact probe-kernel quotienting on each domain. Table 5 checks the principal same-scope, formation-failed, hidden-load, and changed-target routes, while deliberately retaining response rivals for W9.10. No fibre cardinality, live-residue value, or assignment orbit is used. $\square$

6. Thirteen independently tested fibre-finality theorems

6.1 Broad pre-finality domains

For each i , first form the testable, positively extendible set

$$B_{9,i} := \{x \in \text{Cand}_{9,i} : \text{Disp}_{9,i}(x) = \text{TESTABLE}, \exists c \in \mathfrak{C}_9^0 \text{ with } \pi_i(c) = x\}. \quad (25)$$

Eligibility and antecedent fidelity place $B_{9,i}$ inside $\mathfrak{F}_i(\theta_i)$, but they do not fix its augmented-probe value. Before the role theorem is applied, the quotient

$$\mathcal{Q}_{9,i}^{\text{pre}} := B_{9,i} / (S_{9,i} \cap (B_{9,i} \times B_{9,i})) \quad (26)$$

has no prescribed cardinality. The canonical certificate proves only that it is nonempty.

For $x \in B_{9,i}$ let

$$\omega_i(x) \in \Omega_i$$

be the response actually carried by x . This is the primary-response factor of $\Theta_i(x)$, not the canonical value ω_i^* . Let $\mathsf{T}_i(x)$ mean that the *input-only applicability conditions* of the role-specific source theorem hold for the antecedent over which x lies. In particular, T_i fixes the source, branch, carrier, initial data, domain, boundary, parameters, and burden needed to apply that theorem, but it does not say that the response carried by x satisfies the theorem’s equations.

The predicates are fixed by the following input ledger. Each row names only data available before the candidate response is inspected.

Table 6: Input-only applicability predicates T_i .

Fibre	Fixed input conditions
F1	One fixed nondegenerate comparison act, comparison scope and report protocol.
F2	The smooth metric base (M, g) , tangent carrier, standard local first-jet comparison type, regularity and no-extra-selector burden.
F3	The attachment-certified structured source, Cauchy data, gauge, orientation, regularity interval and failure surface.
F4	The metric–scalar Cauchy carrier, attached source, regular Legendre branch, conventions and compact-boundaryless input.
F5	The source-preserving proper action groupoid, regular slice input and declared charge/large/-sector guards.
F6	The Branch-A real-scalar core and the declared absence of a WDW-4B or internal carrier.
F7	The full-regular constraint map, attached source, clock patch, plaque domain, boundary and failure input.
F8	The Branch-A carrier, proper quotient, chart, test algebra, bosonic polarization, smoothing data, CCR normalization, grammar and common domain.
F9	The accepted ordering, heat-regulator grammar, finite-normalization profile, candidate syntax, source, cap/trace-core data and regulator-free endpoint burden; no counterterm coefficient is fixed.
F10	The constructed successor domain, Branch-A zero-theta target and the declared stronger-claim exclusions; no bracket or anomaly value is fixed.

Fibre	Fixed input conditions
F11	The authenticated pointed endpoint, fixed nonzero normalized joint solution, scale-profile input and continuation-interface types; no form, radical, quotient or represented algebra is fixed.
F12	The accepted WDW-6 physical line, attached source history, regular clock window and no-spectator burden; no conditional-dynamics response is fixed.
F13	The literal WDW-8 handoff, fixed source, single-rate comparison carrier, generated-candidate grammar, recovery domain, failure register and nonclaim scope; no lawful recovery response is fixed.

That missing candidate-to-theorem bridge is supplied explicitly. For each role define a response-realization relation

$$R_i \subseteq \{\alpha_i\} \times \Omega_i, \quad R_i(\alpha_i, \omega_i(x)), \quad (27)$$

whose clauses are listed in Table 7. The relation asserts that the response carried by x solves the source equations, realizes the source universal property, or makes the source construction diagrams commute. No R_i contains x_i^* , ω_i^* , $S_{9,i}$, or equality under $\widehat{\Theta}_{9,i}$. The clauses are extracted from the displayed definitions and theorem domains of the accepted predecessors and are fixed before any element of $B_{9,i}$ is tested. They add no constructor to the W9.8 grammar.

Table 7: Candidate-specific response-realization relations.

Fibre	Law asserted of the response carried by x	Source theorem acting on every realizer
F1	The carried governance record instantiates Adm, Standing, Reference and Irreversibility on the same nondegenerate act, and its evaluated-use, quotient, continuation, no-repair and report squares commute.	Kernel necessity; Fixed-Domain Theorems 7.1–7.4; Same-Scope Operator Exhaustion Theorem 7.1.
F2	The carried ∇' occupies the declared MinMetric-Nat1 class on the fixed (M, g) ; $C = \nabla' - \nabla^{LC}$ has the stated local, first-jet, natural, finite-linear, metric, torsion-free and no-selector laws; derivative, curvature and holonomy coordinates are functorially computed from ∇' .	WDW-1 Lemma 7.3, Theorem W1.1 and Proposition C.1.
F3	The carried development solves the attached Einstein–Klein–Gordon Cauchy problem on the fixed data and interval; its source projections, attachment witness, Ward/exchange equations and split/reconstruction maps commute.	Geometric Cauchy uniqueness at the imported regularity; WDW-1 W1.2 and W1.5.
F4	The carried canonical package is obtained by the displayed ADM split and Legendre equations, satisfies the geometric constraint and multiplier equations, reconstructs the same attached source history, and obeys the core boundary differentiability law.	WDW-1 W1.3–W1.7 and the split/reconstruction inverse identities of W1.5.
F5	The carried object satisfies the proper-action slice and characteristic reduction laws, together with the universal descent squares for the source, symplectic form, observables and normal family, while retaining every residual sector coordinate.	WDW-2 W2.4–W2.8.
F6	The carried internal-label family has no inhabitants and its Gauss, representation and incidence arrows have empty domain and the declared neutral codomains.	Empty-family universal property and the Branch-A clauses of WDW-1–5.

Fibre	Law asserted of the response carried by x	Source theorem acting on every realizer
F7	The carried distribution is the image of the constraint/conormal anchor and equals the characteristic kernel; its evaluated plaque, path/homotopy functor and finite-route comparison satisfy the source, clock, boundary and failure diagrams.	WDW-3 W3.4–W3.8 and Proposition 36.2.
F8	The carried first and second differentials satisfy the printed Bastiani formulas and estimates; its regulated contraction, principal symbol, CCR action, common domain, transpose solution squares and complete matching obey the WDW-5 construction laws.	WDW-5 W5.2–W5.13.
F9	The carried counterterm column satisfies $M_\infty(\epsilon)b_{\epsilon,b,\eta}^{\text{ct}}[N] + (h_{\leq 0,\epsilon,b,\eta}[N] - \nu_{b,\eta}[N]) = 0$ and the actual insertion identity; its capwise operators satisfy the regulator limit, literal-overlap, graph-transport, completion and common-domain laws.	W6R.11 and W6R.S12–W6R.S15, including their relative-uniqueness clauses.
F10	On the constructed successor domain, the carried complete constraint family satisfies every linear, Hamiltonian–linear and Hamiltonian–Hamiltonian bracket equation, and its total anomaly record realizes the declared Branch-A zero-theta section with all component records.	W6R.O16–W6R.O19 and the accepted Gate-III original-contract closure.
F11	From the antecedently fixed normalized joint solution $F_A^{(0)}$, the response satisfies $\Psi_A(u) = u(F_A^{(0)})$, $B_A(u, v) = \overline{\Psi_A(u)}\Psi_A(v)$, $\text{Rad } B_A = \ker \Psi_A$; its completion, scalar representation and eleven matching/interface squares commute.	WDW-6 Theorems W6R.20–W6R.24 and the pointed Branch-A capstone theorem.
F12	The carried clock is a source-generated physical-clock quotient; its trace is the first nonlossy Cauchy doublet, the constraint factors through its companion operator, and the connection, propagator, covariant Hamiltonian and complete record satisfy the WDW-7 equations.	WDW-7 Theorem 8.2, Proposition 16.1, Theorem 16.2 and Theorem 36.1.
F13	The carried package is generated in $\text{Cand}_g^{\text{fix}}$ and obeys the affine-phase/unit-amplitude generation, HJ/transport/reconstruction, RPI, lineage, source-response, process, exchange-cube and five-error laws on the fixed recovery domain.	WDW-8 Theorems 1.3–1.4 and 7.1–7.4.

Write the input-only tests as the ordered family $t_{i,1}, \dots, t_{i,m_i}$ and the displayed realization law as the ordered atomic family $r_{i,1}, \dots, r_{i,n_i}$. For every $x \in B_{9,i}$ define

$$\text{RespDisp}_{9,i}(x) = \begin{cases} \text{FAIL}(t_{i,k}), & k \text{ is the least failed input test,} \\ \text{FAIL}(r_{i,\ell}), & T_i(x) \text{ and } \ell \text{ is the least failed realization clause,} \\ \text{SURVIVES,} & T_i(x) \wedge R_i(\alpha_i, \omega_i(x)). \end{cases} \quad (28)$$

Bivalence makes these cases total and disjoint. A response with correct inputs but an altered coefficient, development, quotient, clock, anomaly, record or recovery package now fails at its own first $r_{i,\ell}$; it is not called a survivor merely because the source theorem is applicable.

Now define the post-test survivor and its correctly restricted quotient:

$$A_{9,i} := \{x \in B_{9,i} : \text{RespDisp}_{9,i}(x) = \text{SURVIVES}\}, \quad (29)$$

$$\mathcal{Q}_{9,i} := A_{9,i} / (S_{9,i} \cap (A_{9,i} \times A_{9,i})), \quad \rho_i := [x_i^*]_{S_{9,i}}. \quad (30)$$

Restriction of an equivalence relation to a subset is automatically an equivalence relation. No saturation claim is needed: the definition of $A_{9,i}$ already requires each of its members to possess its own certificate extension. The accepted predecessor theorems and $c_9^* \in \mathfrak{C}_9^0$ show that $x_i^* \in A_{9,i}$, but do not by themselves show that a second member has the same probe value.

Lemma 6.1 (No output loading and no graph loading). *For every i , membership in $B_{9,i}$, satisfaction of T_i , and the definition of R_i are logically prior to the response equality*

$$\hat{\Theta}_{9,i}(x) = \hat{\Theta}_{9,i}(x_i^*).$$

The realization relation tests source laws rather than canonical output. The equality must be derived by applying the corresponding source uniqueness or exhaustion theorem to a candidate-specific realizer. Thus $\mathcal{Q}_{9,i}^{\text{pre}}$ remains allowed to be plural until 9C.i is proved.

Proof. Definition 25 uses only TESTABLE status and existence of a compatible certificate. TESTABLE uses eligibility, antecedent fidelity and formation, none of which contains the canonical response value. Every middle column of Table 7 is stated as equations, universal properties, solution membership or commuting diagrams on the carried response; none refers to the canonical occupant, its probe value or its skin class. The right-hand column supplies the independently accepted comparison theorem that turns those laws into the equality displayed above. $\square$

Lemma 6.2 (Source comparison fixes incident restrictions). *Suppose the source theorem in row Fi identifies every T_i -applicable R_i -realizer with the accepted primary response class. Then every incident value $q_{e,i}(x)$ also equals the accepted Fi value. This implication uses no finality theorem for the fibre at the other end of e .*

Proof. Each $q_{e,i}$ was fixed in W9.3 as the inherited restriction functor from the Fi response to its common-base type. The source comparison is source-, domain-, boundary- and report-preserving, so naturality sends it to equality in $\mathfrak{B}_e$ (or to equality in the upstream quotient already built into that base). Tuple compatibility then requires the opposite leg to carry the same value, but is not used to prove the Fi value itself. $\square$

6.2 The thirteen response exhaustions

Proposition 6.3 (9C.1 – Kernel/admissibility finality). $\mathcal{Q}_{9,1} = \{\rho_1\}$.

Proof. Let $x \in A_{9,1}$. By (28), the governance record actually carried by x satisfies R_1 on the fixed nondegenerate act; nondegeneracy is supplied independently by the canonical certificate. Kernel necessity therefore instantiates Admissibility, Standing, Reference and Irreversibility for that record. Fixed-domain exhaustion classifies its evaluated use, quotient, continuation and report, while same-scope exhaustion sends a carrier, grammar, domain or operator change to failure or reset. Hence the full $\mathsf{F1}$ response of x equals that of K_0 ; equivalently $x S_{9,1} x_1^*$. Since the latter is occupied, the restricted quotient has exactly the displayed class. No physical packages are identified by this unary governance result. $\square$

Proposition 6.4 (9C.2 – Connection finality). $\mathcal{Q}_{9,2} = \{\rho_2\}$.

Proof. Let $x \in A_{9,2}$ and write its carried connection as ∇' . The input test fixes (M, g) and the comparison carrier; R_2 separately asserts of this ∇' every MinMetricNat1 law in Table 7. Thus the difference from ∇^{LC} is a local first-jet natural $(1, 2)$ -tensor in the exact quantified domain of WDW-1 Lemma 7.3. That lemma and Theorem W1.1 force the tensor to vanish; Proposition C.1 exhausts the enriched alternatives. E01 and E02 then force the same source use and recovered curvature. Every factor of $\hat{\Theta}_{9,2}$ agrees, so $x S_{9,2} x_2^*$. Occupation gives the singleton quotient. $\square$

Proposition 6.5 (9C.3 – Covariant Einstein/source finality). $\mathcal{Q}_{9,3} = \{\rho_3\}$.

Proof. For $x \in A_{9,3}$, $\mathsf{T}_3(x)$ supplies the structured source ancestry, attachment certificate, Cauchy data, gauge, orientation, interval and failure boundary, but not a development. The independent relation R_3 asserts that the development actually carried by x solves that attached EKG Cauchy problem and that its Ward and reconstruction squares commute. Local geometric uniqueness therefore forces one development class. WDW-1 W1.2 and W1.5 retain the complete source fibre and tensor witness; E03, E04 and E19 are the corresponding functorial uses. Thus all $\mathsf{F3}$ response and incidence coordinates agree and $x S_{9,3} x_3^*$. $\square$

Proposition 6.6 (9C.4 – Canonical carrier and boundary finality). $\mathcal{Q}_{9,4} = \{\rho_4\}$.

Proof. Take $x \in A_{9,4}$. Its input datum is fixed by T_4 , while R_4 requires the package actually carried by x to satisfy the ADM split, Legendre, constraint, multiplier, reconstruction and boundary laws. WDW-1 W1.3–W1.6 prove that every such realizer has the displayed phase carrier, symplectic form, lapse/shift multiplier status, Hamiltonian and momentum constraints, density/sign conventions and source-preserving inverse. W1.7 fixes the compact boundary classification. E03, E05, E07 and E20 are computed restrictions of those data. Consequently the F4 augmented response equals the canonical one, so $x S_{9,4} x_4^*$. $\square$

Proposition 6.7 (9C.5 – Proper spatial-descent finality). $\mathcal{Q}_{9,5} = \{\rho_5\}$.

Proof. For $x \in A_{9,5}$, the WDW-2 input hypotheses hold and R_5 says that the quotient and descent maps carried by x realize the proper-action slice, characteristic reduction and every universal factorization square. W2.4 therefore identifies its local characteristic quotient, W2.5–W2.7 identify its global groupoid, source, observable and normal-family descent, and W2.8 collects their universal property. The canonical-interface exhaustion retains improper, charged, large, disconnected, singular and sector-changing residuals. E05, E08 and E09 synchronize the canonical, refoliation and quantum uses. Therefore every F5 probe factor agrees with C_{prop} , and $x S_{9,5} x_5^*$. $\square$

Proposition 6.8 (9C.6 – Empty internal-gauge finality). $\mathcal{Q}_{9,6} = \{\rho_6\}$.

Proof. The F6 input is the Branch-A real-scalar core with no internal carrier, and R_6 requires the candidate's carried internal family to be empty and its Gauss, representation and incidence arrows to have that empty domain. There is exactly one empty indexed family and one map from it to each neutral incidence object. E06 and E10 are therefore those same unique restrictions. A nonempty group, representation, connection, Gauss operator or anomaly datum changes α_6 and is a reset, not a second response. Hence every $x \in A_{9,6}$ is $S_{9,6}$ -equivalent to x_6^* . This proves nothing about a nonempty gauge target. $\square$

Proposition 6.9 (9C.7 – Refoliation/HDA finality). $\mathcal{Q}_{9,7} = \{\rho_7\}$.

Proof. Let $x \in A_{9,7}$. The relation R_7 places the distribution, plaque, evaluation and finite-route maps actually carried by x in the quantified comparison class of WDW-3 W3.4 and W3.5. Those theorems identify the full characteristic distribution and evaluated plaque on the fixed regular constraint/source datum. Proposition 36.2 classifies every finite-route difference: class 1 is proper local refoliation, while classes 2–8 change isotropy, topology, charge, source, record, target or existence data. W3.6–W3.8 retain all those residuals and prove source, observable, boundary and clock coherence. E07, E08, E11 and E18 synchronize the external uses. The only surviving same-antecedent response has the canonical F7 probe, so $x S_{9,7} x_7^*$. $\square$

Proposition 6.10 (9C.8 – Configuration differential and representation finality). $\mathcal{Q}_{9,8} = \{\rho_8\}$.

Proof. For $x \in A_{9,8}$, T_8 fixes the Branch-A carrier, quotient, chart, test algebra, smoothing, CCR normalization, grammar and domain. The distinct R_8 asserts that the differentials, regulated contractions, representation maps, domain actions and solution squares carried by x satisfy the WDW-5 construction laws. W5.2–W5.12 then identify the tangent/cotangent pairing, independent Bastiani differential, principal symbol, representation action and solution functor. W5.13 exhausts every remaining variable coordinate and permits equivalence only after the complete nine-coordinate matching. E09–E12 add the proper, empty-internal, principal-symbol and operator-domain matches. Therefore the complete F8 augmented probe equals the canonical value and $x S_{9,8} x_8^*$. $\square$

Proposition 6.11 (9C.9 – Ordering, regulator, and domain finality). $\mathcal{Q}_{9,9} = \{\rho_9\}$.

Proof. Take $x \in A_{9,9}$ and denote its carried counterterm column by b_x^{ct} . Membership in $A_{9,9}$ does not merely make W6R.11 applicable: R_9 tests the candidate-specific equation

$$M_\infty(\epsilon) b_{x;\epsilon,b,\eta}^{\text{ct}}[N] + (h_{\leq 0,\epsilon,b,\eta}[N] - \nu_{b,\eta}[N]) = 0$$

and the actual insertion identity. The locally finite inverse in W6R.11 therefore forces $b_{x;\epsilon,b,\eta}^{\text{ct}}[N] = -M_\infty(\epsilon)^{-1}(h_{\leq 0,\epsilon,b,\eta}[N] - \nu_{b,\eta}[N])$ and cancels the complete local obstruction. W6R.S12, W6R.S12a and W6R.S13–W6R.S15 construct the capwise operator, literal overlaps, scale-scheme transport, successor graph completion and common domain on the same trace core. The old exact unrenormalized lift is still rejected by its heat-trace witness; an altered ordering, source, finite-normalization profile or domain misses a theorem antecedent or resets the target, while an altered counterterm response fails a displayed $r_{9,\ell}$ realization test rather than surviving on applicability alone. E12 and E13 fix the input grammar and anomaly endpoint. Thus all F9 probe factors agree and $x S_{9,9} x_9^*$. $\square$

Proposition 6.12 (9C.10 – Anomaly and closure finality). $\mathcal{Q}_{9,10} = \{\rho_{10}\}$.

Proof. For $x \in A_{9,10}$, the successor domain is fixed by T_{10} and R_{10} requires the complete linear, Hamiltonian–linear and Hamiltonian–Hamiltonian bracket equations to hold for the constraint family actually carried by x , before its total anomaly record is formed. W6R.O16–W6R.O19 then force the declared zero-theta Branch-A closure. E13 fixes the successor endpoint and E14 fixes the constraint images annihilated by the physical functional. A nonclosed response fails the theorem package; a different theta, branch, temporal/gerbe burden or stronger anomaly claim resets the target. Hence $x S_{9,10} x_{10}^*$. $\square$

Proposition 6.13 (9C.11 – Physical inner-product and observable finality). $\mathcal{Q}_{9,11} = \{\rho_{11}\}$.

Proof. Let $x \in A_{9,11}$. The relation R_{11} requires the response carried by x to be built from the antecedently fixed authenticated normalized joint solution by the candidate-specific law $\Psi_x(u) = u(F_A^{(0)})$; it also requires the form, radical, completion, representation and matching squares printed in Table 7. WDW-6 Theorem W6R.20 identifies the resulting nonzero complete-constraint annihilator. Theorem W6R.21 then gives

$$B_A(u, v) = \overline{u(F_A^{(0)})} v(F_A^{(0)}), \quad \text{Rad } B_A = \ker \Psi_A,$$

and Theorem W6R.22 gives the completed rank-one quotient and represented algebra $\mathbb{C}I$. Theorems W6R.23–W6R.24 identify fixed-profile addresses only after the complete state/form/quotient/algebra matching. E14–E16 preserve the constraint images, conditioning interface, and physical/recovery-component separation. Thus the entire F11 response agrees and $x S_{9,11} x_{11}^*$. $\square$

Proposition 6.14 (9C.12 – Clock and conditional-dynamics finality). $\mathcal{Q}_{9,12} = \{\rho_{12}\}$.

Proof. For $x \in A_{9,12}$, R_{12} states that the clock, trace, companion, connection, propagator and record actually carried by x satisfy the WDW-7 source-generation and factorization laws. Theorem 8.2 then exhausts the physical-clock role and leaves the function-valued source-generated quotient $\phi(\cdot) = \tau \mathbf{1}$ up to orientation-preserving reparametrization. Proposition 16.1 and Theorem 16.2 derive the Cauchy doublet as the first nonlossy trace and eliminate point-only, polarized and enriched substitutes. Factorization through the rank-one WDW-6 coordinate and the no-spectator condition force the generated line; reapplication of the constraint fixes the companion response, Hermitian connection, unitary propagator and zero covariant Hamiltonian, as collected in WDW-7 Theorem 36.1. E15, E17 and E19 retain the form, lineage, record and source history. Therefore $x S_{9,12} x_{12}^*$. $\square$

Proposition 6.15 (9C.13 – Record, continuation, and reconciliation finality). $\mathcal{Q}_{9,13} = \{\rho_{13}\}$.

Proof. Let $x \in A_{9,13}$. WDW-8 Theorem 1.3 and Definition 1.5 fix the comparison domain and broad candidate universe before selecting a response. The relation R_{13} additionally requires the package carried by x to satisfy the generated phase/amplitude, recovery, RPI, lineage, process, exchange-cube and five-error laws; it does not require equality to C_{rec} . Theorems 7.1 and 7.2 give a first-divergence normal form and uniform testing of that very package. Theorem 7.3 closes its five errors, and Theorem 7.4 proves that every such lawful realizer belongs to the unique same-scope class. E02, E04, E16–E18 and E20 synchronize its connection, source, physical anchor, clock/lineage, characteristic and canonical uses. Every F13 augmented-probe factor therefore agrees with C_{rec} , so $x S_{9,13} x_{13}^*$. $\square$

Theorem 6.16 (W9.10 – Thirteen non-output-loaded fibre-finality theorem). *For every $i = 1, \dots, 13$,*

$$\mathcal{Q}_{9,i} = \{\rho_i\}.$$

Each singleton is positively occupied. The conclusion is obtained only after the broad $B_{9,i}$ domain, input-only applicability test, candidate-specific response-realization test, and source uniqueness theorem have been applied in that order. Every member of $B_{9,i} \setminus A_{9,i}$ retains its least failed input or realization test from (28) and is not silently discarded.

Proof. Propositions 6.3–6.15 first use $R_i(\alpha_i, \omega_i(x))$ to place the response actually carried by each $x \in A_{9,i}$ in the quantified solution or universal-property domain of the cited predecessor theorem. Only that theorem then gives the canonical primary-response class; Lemma 6.2 supplies every incident restriction, while the fixed antecedent supplies the local locks. Thus the full augmented-probe value is canonical and the candidate belongs to ρ_i . The canonical source constructions satisfy every R_i , and c_0^* supplies compatible extensions, so $x_i^* \in A_{9,i}$ and the class exists. Lemma 6.1 shows that neither TESTABLE status, antecedent fidelity, certificate extendibility, the realization equations nor the restricted skin relation inserted the equality. A failed realization is recorded; a changed antecedent remains a reset. $\square$

7. Compatibility descent, live residue, and the lawful orbit

7.1 Compatibility descends to fibre skin classes

Lemma 7.1 (Representative-independent compatibility). *Let $x_i, y_i \in A_{9,i}$ with $x_i S_{9,i} y_i$ for every i . Then*

$$\text{Compat}(x_1, \dots, x_{13}) = \text{Compat}(y_1, \dots, y_{13}).$$

If the first tuple is compatible, the second tuple extends to an object of $\mathfrak{C}_9^0$.

Proof. By (18), $S_{9,i}$ preserves every incident value $q_{e,i}$ and every source, domain, boundary, failure, and report lock. Hence each E01–E20 equality and each lock equality has the same truth value after representative replacement. If the first tuple is compatible, its replacement therefore lies in $\mathfrak{J}_9$. W9.5 constructs $L_9(\mathbf{y})$ and its universal cocone directly from that compatible tuple, and the pullback definition (12) then supplies a complete certificate over $\mathbf{y}$. No presentation arrow witnessing a kernel-pair relation is assumed. $\square$

Define the all-survivor certificate subgroupoid

$$\mathfrak{C}_9^{\text{surv}} := \{c \in \mathfrak{C}_9^0 : \pi_i(c) \in A_{9,i} \text{ for every } i\}. \quad (31)$$

Proposition 7.2 (Complete-certificate class finality). *The survivor certificate quotient is nonempty and singleton:*

$$\mathfrak{C}_9^{\text{surv}} / (S_9^{\text{cert}} \cap (\mathfrak{C}_9^{\text{surv}} \times \mathfrak{C}_9^{\text{surv}})) = \{[c_9^*]_{S_9^{\text{cert}}}\}.$$

Proof. The canonical certificate is a member. If $c \in \mathfrak{C}_9^{\text{surv}}$, W9.10 gives $\pi_i(c) S_{9,i} x_i^*$ for every i . Both projected tuples are in $\mathfrak{J}_9$. Proposition 4.9 now identifies their tensor-mediator classes and every complete-probe factor, so $c S_9^{\text{cert}} c_9^*$. The proof uses the normalized interface diagram and colimit uniqueness, not fibre presentation arrows. $\square$

Thus antecedent eligibility/fidelity and compatibility define class-level predicates on the typed quotients. Write $\iota_j : \mathcal{Q}_{9,j} \rightarrow \mathcal{Q}_9 := \bigsqcup_k \mathcal{Q}_{9,k}$ for the typed coproduct injections. To avoid evaluating an F_i predicate on an F_j representative outside its domain, define the global predicates by the coproduct eliminator

$$\text{Elig}_i^{\mathcal{Q}}(\iota_j[x]) := \begin{cases} \text{Elig}_i(x), & j = i, \\ 0, & j \neq i, \end{cases} \quad (32)$$

$$\text{Fid}_i^{\mathcal{Q}}(\iota_j[x]; \theta_i) := \begin{cases} \text{Elig}_i(x) \wedge \text{Fid}_i^{\text{ant}}(x; \theta_i), & j = i, \\ 0, & j \neq i. \end{cases} \quad (33)$$

This case split is the elimination rule for the typed coproduct, not an evaluation of a partial predicate and not a user-supplied role label. The coproduct tags are the carrier types later characterized intrinsically by Sig in (36). For a thirteen-tuple of quotient classes, $\text{Compat}^{\mathcal{Q}}$ is the common truth value of compatibility on any representatives. W9.7 makes the nonzero branches of (32)–(33) representative-independent, and Lemma 7.1 does the same for $\text{Compat}^{\mathcal{Q}}$.

7.2 Residue before canonical identification

The set $\mathcal{Q}_9$ above is the typed admissible response residue. The coproduct tags are carrier types, not arbitrary labels. Define

$$\Sigma_9^{\text{pre}} = \left\{ \sigma : R_9 \hookrightarrow \mathcal{Q}_9 : \text{Elig}_i^{\mathcal{Q}}(\sigma(r_i)) \wedge \text{Fid}_i^{\mathcal{Q}}(\sigma(r_i); \theta_i) \forall i, \quad \text{Compat}^{\mathcal{Q}}(\sigma) = 1 \right\}, \quad (34)$$

and only then

$$\text{Residue}_9^{\text{live}} = \bigcup_{\sigma \in \Sigma_9^{\text{pre}}} \text{im } \sigma. \quad (35)$$

“Live” is therefore a property of quotient classes occurring in at least one compatible injection. It is not a raw-candidate status and is not a synonym for the binary skin relation.

7.3 Structural role signatures

To prevent role names from functioning as a selector, attach to every typed class $[x]_i$ the unlabeled structural signature

$$\text{Sig}([x]_i) = (\text{CarrierType}, \text{DistinguishedMap}, \text{AugmentedProbeCodomain}, \text{IncidenceProfile}, \theta_i). \quad (36)$$

The role signature is the canonical value $\text{Sig}(r_i) := \text{Sig}(\rho_i)$. Every factor is preserved by a certificate-lawful isomorphism.

Table 8: Pairwise separating role signatures.

Role	Carrier invariant	Distinguished structure	Separating probe invariant
F1	Four-place governance profile	Fixed-domain status map	Adm/St/Ref/Irr and report codomain; no physical carrier.
F2	Affine connection	Covariant derivative and curvature	Torsion/metricity/holonomy profile.
F3	Structured Lorentzian solution fibre	Source attachment and Cauchy evolution	Nonvacuum history, Ward/exchange, source witness.
F4	Weakly symplectic ADM phase carrier	Hamiltonian/momentum moment map	Canonical density/sign and compact boundary ledger.
F5	Proper action groupoid	Orbit/slice quotient morphism	Proper, improper, large, and charged separation.
F6	Empty indexed family	Unique empty incidence	Cardinality-zero internal label invariant.
F7	Conormal Lie algebroid and path groupoid	HDA anchor and evaluated characteristic	Refoliation/plaque/failure profile.
F8	Bastiani function algebra	First/second configuration differential	CCR seed, principal symbol and syntax profile.
F9	Successor graph-topological carrier	Regulator/counterterm endpoint map	Ordering, graph domain and completion profile.
F10	Closed constraint/anomaly system	Total anomaly/closure map	Zero-theta Branch-A closure record.
F11	Positive Hermitian quotient	Terminal functional and radical quotient	Nonzero rank one and represented $\mathbb{C}I$.
F12	Clock-indexed Hermitian line bundle	Conditioning, connection and propagator	Function-valued clock, $H_{\text{rel}} = 0$, record use.
F13	Moore-history/process/cube package	RPI continuation and five-error comparison	Disjoint recovery carrier, exchange cube, error/failure vector.

Lemma 7.3 (No cross-role eligibility). *For the thirteen canonical residue classes,*

$$\text{Elig}_i^{\mathcal{Q}}(\rho_j) \wedge \text{Fid}_i^{\mathcal{Q}}(\rho_j; \theta_i) \iff i = j.$$

Proof. If $j \neq i$, the coproduct eliminator (32)–(33) returns zero without evaluating an Fi predicate on the Fj representative. Intrinsically, the carrier and distinguished-map columns of Table 8 are pairwise nonisomorphic. Where a generic carrier category could support more than one listed structure, the augmented probe, incidence profile, and antecedent parameter separate them. Thus a lawful certificate isomorphism cannot erase the typed coproduct distinction. For $j = i$, W9.10 and positive occupation by c_9^* give the reverse implication. $\square$

Theorem 7.4 (W9.11 – Live-residue and assignment-space theorem). *The compatible pre-assignment space is nonempty and*

$$\Sigma_9^{\text{pre}} = \{\sigma_\star\}, \quad \sigma_\star(r_i) = \rho_i.$$

Consequently

$$\text{Residue}_9^{\text{live}} = \{\rho_1, \dots, \rho_{13}\}, \quad |\text{Residue}_9^{\text{live}}| = 13,$$

and $\sigma_\star$ is a compatible bijection onto the independently formed live residue.

Proof. The canonical certificate (14) proves that the canonical classes are pointwise eligible, faithful, and compatible. Representative independence is Lemma 7.1, so $\sigma_* \in \Sigma_9^{\text{pre}}$. Let σ be any member. W9.10 places its value at r_i in some singleton $\mathcal{Q}_{9,j} = \{\rho_j\}$. Pointwise structural eligibility and Lemma 7.3 force $j = i$. Thus $\sigma(r_i) = \rho_i$ for every i . Substitution in (35) gives the displayed residue; its thirteen members remain distinct because the coproduct carrier types and signatures are distinct. $\square$

7.4 Automorphisms of the actual complete certificate

Definition 7.5 (Lawful automorphism group). Γ_9^{law} is the automorphism group of the complete-certificate presentation groupoid $\mathfrak{C}_9^0$ preserving:

1. the projection to $\mathfrak{J}_9$, the ledger projection to $\mathfrak{L}_9$, the universal cocone, and the witness map to $\widehat{W}_9$;
2. Θ_9^{ten} , Θ_9^{cert} , S_9^{cert} , and every $S_{9,i}$;
3. the structural signatures, frozen antecedents, the broad domains $B_{9,i}$, the response projections ω_i , realization relations R_i , response-test dispositions (28), and survivor subsets $A_{9,i}$;
4. E01–E20 and all composite incidence equalities;
5. source ancestry, domains, boundary, failure, and report coordinates;
6. the physical-anchor/recovery-component distinction and the typed empty internal-gauge coordinate.

An actual internal presentation arrow acts trivially on $\mathcal{Q}_{9,i}$ when its endpoints lie in $S_{9,i}$. The kernel pair may also relate probe-identical objects for which no presentation arrow was registered; the quotient and compatibility proof use the relation itself, not an invented arrow.

Proposition 7.6 (Automorphism action). Γ_9^{law} acts on

$$\Sigma_9 := \{\text{compatible bijections } R_9 \rightarrow \text{Residue}_9^{\text{live}}\},$$

and every lawful automorphism fixes every ρ_i .

Proof. Preservation of the certificate projections, fibre skins, eligibility, antecedent fidelity, response-realization relations, response-test dispositions, survivor subsets, and incidence values makes the action on quotient classes and compatible bijections well defined. A lawful automorphism preserves $\text{Sig}(r_i)$. Pairwise separation in Table 8 prevents motion from the Fi residue to Fj for $j \neq i$, and W9.10 leaves only one Fi class. Therefore every ρ_i is fixed. The argument uses actual certificate structure, not printed role names. $\square$

Theorem 7.7 (W9.12 – Lawful automorphism and assignment-orbit theorem). *The final assignment space is $\Sigma_9 = \{\sigma_*\}$ and*

$$|\Sigma_9 / \Gamma_9^{\text{law}}| = 1.$$

Proof. W9.11 identifies the live residue and its only pointwise eligible compatible bijection. Proposition 7.6 shows that every lawful automorphism fixes it. The quotient therefore has one orbit. $\square$

Remark 7.8 (Why cardinality does not create occupation). The number thirteen was not used to infer a bijection. W9.4 populated the joint tuple, W9.6 populated the complete certificate, W9.10 proved fibre finality in correctly typed quotients, and W9.11 formed the live residue from all representative-independent compatible injections. Only afterward was the orbit counted.

8. Physical quotient, time and record standing, and WDW lineage

8.1 The physical object inside the joint certificate

Let $\mathcal{I}_A^{\text{ren}}$ be the complete Branch-A renormalized constraint-label family. It contains the proper momentum labels and renormalized Hamiltonian labels; the internal Gauss subfamily is empty. On the accepted common graph source, the joint solution space is

$$\mathcal{Z}_A = \bigcap_{\alpha \in \mathcal{I}_A^{\text{ren}}} \ker \widehat{C}_\alpha^\times, \quad F_A^{(0)} \in \mathcal{Z}_A \setminus \{0\}, \quad F_A^{(0)}(q_*) = 1.$$

The empty Gauss intersection contributes the ambient space and no additional operator equation.

On the renormalized test/source dual $\mathcal{E}_A^{\text{ren}}$, define

$$\Psi_A(u) := u(F_A^{(0)}), \quad B_A(u, v) := \overline{\Psi_A(u)}\Psi_A(v).$$

The WDW-6 theorem gives

$$\text{Rad } B_A = \ker \Psi_A, \quad \mathcal{H}_{\text{mathrmphys},9} := \overline{\mathcal{E}_A^{\text{ren}} / \text{Rad } B_A} \xrightarrow[u]{\mapsto} \Psi_A(u) \cong \mathbb{C}. \quad (37)$$

Scalar multiplication descends to the faithful represented algebra

$$\mathfrak{A}_{\text{mathrmphys},9} = \mathbb{C}I.$$

Theorem 8.1 (W9.13 – Physical joint-kernel and quotient identification). *The F11 coordinate of Ξ_9^* places the normalized nonzero state, complete joint annihilator, positive form, radical, quotient (37), and represented algebra inside the completed joint certificate. This physical object is compatible with every other fibre and is unique up to lawful skin at the frozen target.*

The WDW-8 elements on $(0, \delta_0]_{\text{rec}}$ are comparison objects and are not additional points of $\mathcal{H}_{\text{phys},9}$.

Proof. E14 identifies the complete anomaly-closed constraint system with the one annihilating $F_A^{(0)}$ and whose images are null in B_A . E15 identifies the WDW-7 generated line as the isometric image of (37); E16 retains the exact physical anchor while keeping the recovery component disjoint. Theorem 3.2 proves global compatibility, and Proposition 6.13 proves target-relative finality. The tensor colimit and either correctly typed skin cannot change this conclusion because the state, form, radical, quotient, algebra, and carrier tag are factors of $\hat{\Theta}_{9,11}$ and therefore of Θ_9^{cert} on complete certificates. $\square$

8.2 Four temporal notions

The joint certificate distinguishes four objects often denoted by the same word “time.”

1. **Bulk presentation data.** A foliation label, coordinate time, lapse, shift and gauge fixing specify a canonical presentation. Proper changes among them are skin when all source, boundary, clock, record and failure probes are preserved.
2. **Physical local clock.** The function-valued scalar quotient $\phi(\cdot) = \tau \mathbf{1}$ is source-generated and transverse on the certified local domain. It labels conditional intersections of one occupied constrained history and carries a failure boundary.
3. **Complete relational record.** On the occupied orbit

$$s_\tau = \Gamma_\star \cap \{\phi(\cdot) = \tau \mathbf{1}\}_{\text{prop}}, \quad V^{\text{rec}}(\tau) = \text{Vol}(h(s_\tau)),$$

and V^{rec} is nonconstant. Record change is physical relational change even though the physical ray is stationary.

4. **Boundary/asymptotic time.** A boundary generator can carry standing on a target with the appropriate boundary, differentiability and charge data. Such data are absent because $\partial\Sigma = \emptyset$ in the core.

The WDW-7 conditioned line has the constraint-induced Hermitian connection ∇_τ^C and

$$i\hbar \nabla_\tau^C \xi_\tau = H_{\text{rel}}^C(\tau) \xi_\tau, \quad H_{\text{rel}}^C = 0.$$

For a parallel section the coefficient in the transported physical line is constant, so the projective ray does not change. The record is a different typed map on the underlying source history and need not be constant.

Theorem 8.2 (W9.14 – Time, boundary, clock, and record standing). *At the frozen target:*

1. *bulk coordinate time, lapse, shift, gauge fixing and foliation have no independent physical standing when the complete probe is unchanged;*
2. *the local function-valued profile clock and complete volume record are standing-bearing on their licensed source-attached use domain;*
3. *$H_{\text{rel}}^C = 0$ and stationary ray transport do not imply absence of relational change, because V^{rec} is nonconstant; and*
4. *boundary/asymptotic time is absent from this compact target, not denied on every enriched target.*

Proof. WDW-3 proves that proper refoliations are characteristic presentation arrows and that no same-target foliation selector survives. WDW-1 proves multiplier status for lapse and shift and separates compact-core boundary absence from boundary-generator extensions. WDW-4A and the physical-clock quotient fix the function-valued clock and its use/failure data. WDW-7 proves quotient-native conditioning, zero line Hamiltonian, stationary ray classification, and nonconstant complete record; E17 and E19 place those objects on the same actual source lineage. The complete joint probe reads the clock, record, source, and boundary coordinates. Hence a complete-certificate presentation change lies in S_9^{cert} exactly when all these values are fixed; its F7, F12, and F13 projections lie in their respective fibre skins. A changed clock, record, or boundary generator is in none of those relations. $\square$

8.3 Constraint-native realization of the WDW equation

The classical constraint map at F4 contains

$$\mathcal{H}_i = 0, \quad \mathcal{H}_\perp = 0,$$

with the exact WDW-1 DeWitt normalization and the WDW-4A scalar source terms. WDW-5 constructs a configuration-space differential calculus and the regulated Hamiltonian candidates on a common algebraic domain. WDW-6 does not merely remove the regulator from the old formula: it excludes that route and constructs a distinct counterterm-enriched successor domain on which the renormalized constraints and anomaly record close and admit $F_A^{(0)}$.

The complete downstream chain is therefore

$$\begin{aligned} & \text{nondegenerate target and standing-preserving connection} \\ & \implies \text{source-attached Einstein–Klein–Gordon object} \\ & \implies \text{ADM constraints, proper descent, and refoliation closure} \\ & \implies \text{regulated syntax, successor domain, and anomaly closure} \\ & \implies \text{nonzero joint solution and positive physical quotient} \\ & \implies \text{profile-clock conditioning, record, and controlled recovery.} \end{aligned} \tag{38}$$

Theorem 8.3 (W9.15 – Constraint-native Wheeler–DeWitt lineage). *On $\mathcal{D}_{\text{WDW}}^{\text{core}}$, the Wheeler–DeWitt equation is an occupied downstream constraint-native realization in the chain (38). It is not merely the formal substitution $\pi^{ij} \mapsto -i\hbar\delta/\delta h_{ij}$, and it is not an upstream principle from which the covariant source, kernel, or connection is inferred.*

The theorem does not assert that AASC alone selects the accepted ordering, regulator, counterterms, couplings or numerical constants.

Proof. F2–F4 and E01–E07 place the covariant, canonical and characteristic objects on one source-attached branch. F5–F10 and E08–E14 place the configuration calculus, regulated syntax, successor domain and complete anomaly closure on that branch. Theorem 8.1 supplies an actual nonzero joint solution and positive quotient. F12–F13 and E15–E20 add the exact clock, record and recovery objects without identifying their carriers. The joint-section theorem and the populated object c_9^* prove that all arrows in (38) coexist in one complete certificate, while Theorem 6.16 and Theorem 7.7 prove complete target-relative occupation. The formal substitution alone supplies none of the domain, regulator, anomaly, state, quotient, clock, or recovery steps and therefore cannot replace the chain. Ordering, regulator grammar and the finite-normalization profile enter through the frozen F9 antecedent; the local counterterm coefficients are then derived by the WDW-6 inverse. None of them comes from the kernel theorem. $\square$

9. Failure boundary and complete realization closure

9.1 Fail-closed domain

Let Fail_k be the complete failure object exported by predecessor G_k . The WDW-9 failure object is

$$\text{Fail}_9 = \bigcup_{k=0}^8 \text{Fail}_k \cup F_{\text{fibre}} \cup F_{\text{overlap}} \cup F_{\text{completion}} \cup F_{\text{probe}} \cup F_{\text{skin}} \cup F_{\text{assignment}}. \tag{39}$$

The new terms have the following exact meanings:

- F_{fibre} : loss of role eligibility, antecedent fidelity, formation, or one independently tested response theorem in F1–F13;

- F_{overlap} : the first noncommuting edge among E01–E20;
- $F_{\text{completion}}$: a missing primary generator, illegal path, extra identification, non-value-reflecting witness, assembly mismatch, or failure of the colimit mediator;
- F_{probe} : an ill-typed tensor/certificate probe, a missing certificate projection, or a standing-bearing distinction not read by the complete probe;
- F_{skin} : domain confusion, failure of congruence or maximality, or a fibre/certificate kernel pair that erases a probe value; and
- $F_{\text{assignment}}$: empty compatible assignment space, an unclassified live residue, failure of bijectivity, or an incorrectly computed automorphism orbit.

The inherited union includes loss of the regular metric branch, source attachment, compact boundary condition, canonical reconstruction, proper quotient, full characteristic regularity, function-valued clock transversality, common operator or graph domain, successor closure, anomaly closure, nonzero joint state, positive form, Cauchy-doublet conditioning, complete record, Hamilton–Jacobi chart, amplitude positivity, homogeneous response chart, evaluated-plaque seed, actual-lineage identity, target-process completeness, upstream exchange ledger, history-indexed cube, compact recovery tube, or any stated error bound.

Theorem 9.1 (W9.16 – No backflow and complete failure boundary). *The WDW-9 theorem is valid only on the complement of (39). At the first failed predecessor, fibre, overlap, completion, probe, skin, or assignment condition, the affected lineage stops with its exact failure status. No later WDW-9 object repairs or extends that lineage as the same target.*

Proof. Proposition 1.3 proves the one-way theorem-use order. The candidate status theorem assigns every new failure a first decisive gate, and Proposition 4.2 prevents completion from erasing an overlap mismatch. Complete-probe and maximal-skin theorems prevent quotient repair of a standing-bearing difference. The residue and orbit constructions require positive occupants and therefore cannot manufacture one after a failure. Taking the union with every imported failure object gives (39) and the stated fail-closed rule. $\square$

9.2 Certificate completeness and realization licensure

Let C_9 denote the certificate containing the frozen target, the populated joint section, L_9^0 , the populated complete-certificate groupoid $\mathfrak{C}_9^0$, Θ_9^{ten} , Θ_9^{cert} , S_9^{cert} , all thirteen $S_{9,i}$, the staged candidate/status maps, the compatibility-descent witness, thirteen fibre residues, live residue, assignment space, failure object, and report ledger. By construction it is:

- candidate-complete by Theorem 5.3;
- quotient-complete by Theorem 4.7;
- role-set complete by Theorem 2.2;
- eligibility-, antecedent-fidelity-, compatibility- and formation-failure complete by Theorem 5.6, with response finality supplied only by Theorem 6.16;
- fibre- and live-residue-complete by Theorems 6.16 and 7.4; and
- assignment-orbit complete by Theorem 7.7.

Theorem 9.2 (W9.17 – Complete joint realization closure). *The certificate C_9 licenses one complete constraint-native realization class of $\mathcal{D}_{\text{WDW}}^{\text{core}}$, unique up to the lawful complete-certificate equivalence S_9^{cert} :*

$$\text{RealizationLicensed}(C_9, \mathcal{D}_{\text{WDW}}^{\text{core}}), \quad |\Sigma_9/\Gamma_9^{\text{law}}| = 1.$$

No changed-target object is identified with that class.

Proof. Theorem 3.2 supplies a nondegenerate joint occupant, and W9.6 supplies the populated complete certificate $\mathfrak{C}_9^*$. The preceding completeness list establishes every antecedent of the accepted role-occupancy theorem: the forced role set is independent, the live residue is formed from all compatible injections, the canonical assignment is a compatible bijection onto it after representative-independent compatibility has been proved, Proposition 7.2 identifies the single complete-certificate survivor class, every failure is classified, and the lawful assignment quotient has one orbit. Applying the sufficiency direction of that theorem yields realization licensure. Target-reset objects were excluded from the same-target residue by Theorem 5.6, so the conclusion is exactly certificate-relative. $\square$

9.3 Endpoint payload and decision

Define

$$G_9 := (\text{Scope}_9, \text{Imports}_9, \text{Fibres}_9, \text{Overlaps}_9, \text{JointPop}_9, \text{Completion}_9, \\ \text{Probe}_9, \text{Skin}_9, \text{Candidates}_9, \text{Residue}_9, \text{Assignments}_9, \\ \text{Physical}_9, \text{TimeRecord}_9, \text{Lineage}_9, \text{Boundary}_9, \text{Cert}_9). \quad (40)$$

The close predicate is

$$\text{Close}(G_9) \iff \left(\bigwedge_{k=1}^{18} G_{9-k} \right) \wedge (\mathfrak{J}_9 \neq \emptyset) \wedge (\Sigma_9 \neq \emptyset). \quad (41)$$

The orbit cardinality is an output of (41); it is not inserted into the antecedent.

Theorem 9.3 (W9.18 – G_9 endpoint and core-ARC decision). *For $\mathcal{D}_{\text{WDW}}^{\text{core}}$,*

$$\mathfrak{J}_9 \neq \emptyset, \quad |\text{Residue}_9^{\text{live}}| = 13, \quad |\Sigma_9/\Gamma_9^{\text{law}}| = 1, \quad G_9 = \text{PASS}.$$

The core WDW ARC therefore closes at one complete lawful realization orbit. Every conclusion is restricted by Theorem 9.1 and the nonclaims below.

Proof. Joint nonemptiness is Theorem 3.2. The live-residue cardinality is Theorem 7.4; the orbit cardinality is Theorem 7.7. Theorems 8.1–8.3 identify the physical, time/record and lineage coordinates, while Theorem 9.1 supplies the complete boundary and Theorem 9.2 supplies licensure. Section 11 matches these results to G9-01–G9-18 with no open gate. Hence (41) holds and the payload (40) has status PASS. $\square$

9.4 Reportability matrix

Table 9: Exact report boundary.

Claim	WDW-9 status
Kernel governance applies to the determinate capstone target	Imported and instantiated; it does not populate a physical fibre.
Accepted WDW-1–8 occupants form one compatible joint tuple and one licensed realization	Compatibility is W9.3–W9.4; certificate assembly and licensure are W9.6 and W9.17, not imported from PASS labels.
The core Gauss family is empty	Exact Branch-A target statement; no Standard-Model closure follows.
The physical state is in the complete constraint kernel and positive quotient	Proved at the authenticated pointed Branch-A target and placed by W9.13.
Bulk coordinate time is primitive	Denied at the fixed complete-probe target.
The zero line Hamiltonian implies no relational change	Denied; the complete volume record is nonconstant.
Boundary time never has standing	Not claimed; the core is boundaryless.
The WDW equation is only formal ADM momentum substitution	Denied by the exact downstream lineage theorem.
AASC alone fixes ordering, counterterms, couplings or constants	Not claimed.
The result is unique across quantum gravity	Not claimed; uniqueness is target- and certificate-relative.
The core result includes WDW-4B/Standard-Model quantum gravity	Not claimed; that is a separate target enrichment.
The result proves a global clock, ultraviolet completion or singularity resolution	Not claimed.
The result proves decoherence, an arrow of time, cosmological selection or phenomenology	Not claimed.
The result closes one complete realization of the frozen core	Proved, with one lawful assignment orbit.

Boundary-complete interpretation. WDW-9 closes the thirteen-fibre certificate of one populated compact-hyperbolic metric–real-scalar target. Deleting the Branch-A, source-attached, local-clock, common-domain, counterterm-successor, zero-theta, pointed physical-quotient, single-rate, no-spectator, disjoint-recovery, strict-cube, or fail-closed qualifications changes the theorem. Adding gauge/fermion, boundary/asymptotic, global, singular, ultraviolet, scattering or phenomenological coordinates requires a new target and positive construction.

10. Certified quantum-standing closure, occupied export, and downstream boundary

The core endpoint W9.18 (Theorem 9.3) closes the frozen WDW realization before any use of the present section. This section fixes a raw representative-level predictive certificate, derives the standing quotient and its naturality, and then constructs one occupied WDW model. Nothing below is an antecedent of W9.1–W9.18.

10.1 Certified representative interface and structural standing closure

Definition 10.1 (Certified **QP/QS** boundary). Let $\mathfrak{S}$ be a nondegenerate determinate closed quantum-predictive theorem object over a domain U . A *representative predictive certificate*

$$c_{\mathfrak{S}}^{\text{QP}} = (X, \Gamma \curvearrowright X, p, \mathcal{C}, \boxtimes, \mathbf{1}, C; \zeta_{\text{prob}}, \zeta_{\text{ph}}, \zeta_{\text{ray}}, \zeta_{\text{sep}}, \zeta_{\text{comp}}, \zeta_{\text{cont}}) \quad (\text{QP-cert})$$

contains a nonempty presentation object X , the lawful scalar-redescription action $\Gamma = \mathbb{C}^\times$, a raw transition-weight family $p : X \times_U X \rightarrow [0, 1]$, a declared comparison class $\mathcal{C}$ with symmetric monoidal product, and a raw closed continuation C . Its six proof coordinates have the following exact burdens:

QP-Prob ζ_{prob} certifies that p is a standing-bearing representative-level comparison family and is stable under the declared base restrictions and lawful arrows;

QP-Phase ζ_{ph} certifies that the Γ -action is lawful redescription and that $p(\gamma x, \gamma' y) = p(x, y)$; it does *not* declare an orbit to be a standing-bearing state;

QP-Ray ζ_{ray} supplies the orbit coequalizer $q : X \rightarrow X/\Gamma$ in the declared regularity class and proves its stability under base restriction; it does *not* assert standing descent;

QP-Sep ζ_{sep} is the raw profile-completeness certificate

$$[p(x, z) = p(y, z) \text{ and } p(z, x) = p(z, y) \ \forall z] \iff q(x) = q(y);$$

QP-Comp ζ_{comp} certifies on representatives that $\boxtimes$ is independently equivariant in both inputs: for every $(\gamma, \gamma') \in \Gamma \times \Gamma$,

$$\boxtimes(\gamma x, \gamma' y) = (\gamma \gamma') \boxtimes(x, y).$$

Equivalently, it is $(\Gamma \times \Gamma)$ -equivariant when the target Γ -action is pulled back through scalar multiplication $\Gamma \times \Gamma \rightarrow \Gamma$. The same certificate proves that transition weights multiply and that the associator, unitors, and symmetry obey their coherence laws; and

QP-Cont ζ_{cont} certifies that C is Γ -equivariant, preserves p , is monoidal on the declared comparison class, and commutes with restriction and lawful arrows.

Write $\mathbf{QP}(\mathfrak{S}; c_{\mathfrak{S}}^{\text{QP}})$ when these raw data and proofs are supplied. In particular, **QP** does not assume that X/Γ carries state standing, does not supply a descended transition weight or continuation, and does not assume quotient-level composition naturality.

A *structural quantum-standing certificate*

$$c_{\mathfrak{S}}^{\text{QS}} = (P, q, \bar{p}, \bar{C}, \bar{\boxtimes}; \delta_{\text{stand}}, \delta_{\text{sep}}, \delta_{\text{cont}}, \delta_{\text{comp}}, \delta_{\text{nat}}) \quad (\text{QS-cert})$$

proves that $P = X/\Gamma$ is the standing-bearing state object; q is the universal standing descent and has exactly the scalar-redescription orbits as fibres; $\bar{p}$ is well defined and separates standing distinctions; $\bar{C}$ is the unique standing-preserving continuation with $\bar{C}q = qC$; $\bar{\boxtimes}$ is the induced strong symmetric-monoidal independent-composition law; and all four constructions are natural under lawful arrows and base restriction. Write $\mathbf{QS}(\mathfrak{S}; c_{\mathfrak{S}}^{\text{QS}})$ when this derived certificate is supplied.

The boundary contains no flat spatial carrier, carrier dimension, Laplacian, preferred time, Wigner–Stone lift, Hamiltonian-response predicate, potential, or scale anchor. Those remain projection-side burdens.

For functorial export, write $\mathbb{C}_U := U \times \mathbb{C}$ for the explicitly defined trivial Hermitian line over U , with its canonical unit section and connection.

Let $\mathbf{QPred}(U)$ be the category of representative predictive models equipped with the complete certificate $c^{\mathbf{QP}}$ of Definition 10.1. An object contains the representative carrier, scalar-redescription action, raw transition weight, declared comparison groupoid, raw continuation, record and first-failure boundary, together with the six proof coordinates $\zeta_{\text{prob}}, \dots, \zeta_{\text{cont}}$. An arrow contains a physical map T , its equivariance square for the scalar action, the algebra, continuation, record and failure maps, and a strong monoidal comparison functor preserving every ζ -certificate. Its object components are $T^{\widehat{\otimes}^n}$, with $T^{\widehat{\otimes}^0} = \text{id}_{\mathbb{C}_U}$, and its action on every internal comparison-groupoid arrow $S : \mathcal{H}^{\widehat{\otimes}^n} \rightarrow \mathcal{H}^{\widehat{\otimes}^m}$ is

$$\mathcal{L}(T)(S) := T^{\widehat{\otimes}^m} S (T^{\widehat{\otimes}^n})^{-1}.$$

Restrictions retain the raw data and restrict each proof coordinate.

Let $\mathbf{QStand}(U)$ be the category of objects of $\mathbf{QPred}(U)$ equipped additionally with the complete derived certificate $c^{\mathbf{QS}}$ of Definition 10.1. An arrow must preserve the quotient map, descended transition weight, descended continuation, induced monoidal product, standing descent, separation, and every arrow/restriction naturality square. Thus membership in $\mathbf{QStand}(U)$ is certified mathematical data, not a consequence of rank, positivity, unitarity, or the types of the record fields alone. Composition and identities are componentwise, including the displayed internal-arrow action.

10.2 Kernel-governed certified standing descent

Proposition 10.2 (Certified quotient closure and naturality). *Let $\mathfrak{S}$ be a nondegenerate determinate admissible quantum predictive interior over U , and suppose only the representative-level certificate*

$$\mathbf{QP}(\mathfrak{S}; c_{\mathfrak{S}}^{\mathbf{QP}})$$

of Definition 10.1. Then the following data are conclusions:

1. *the orbit coequalizer $P_{\mathfrak{S}} = X_{\mathfrak{S}}/\mathbb{C}^{\times}$ is the standing-bearing state object and $q_{\mathfrak{S}} : X_{\mathfrak{S}} \rightarrow P_{\mathfrak{S}}$ is the universal standing descent;*
2. *there is a unique separated transition weight*

$$\bar{p}(qx, qy) = p(x, y);$$

3. *there is a unique standing-preserving continuation $\bar{C}(qx) = q(Cx)$; and*
4. *representative composition descends uniquely to a strong symmetric monoidal product*

$$q(x \boxtimes y) = qx \boxtimes qy,$$

and all three descended constructions are natural under lawful arrows and base restriction.

These conclusions form a certificate $c_{\mathfrak{S}}^{\mathbf{QS}}$ and hence

$$\mathbf{QP}(\mathfrak{S}; c_{\mathfrak{S}}^{\mathbf{QP}}) \implies \mathbf{QS}(\mathfrak{S}; c_{\mathfrak{S}}^{\mathbf{QS}}). \quad (42)$$

The construction is functorial:

$$\mathbf{QClose}_U : \mathbf{QPred}(U) \longrightarrow \mathbf{QStand}(U), \quad (\mathfrak{S}, c^{\mathbf{QP}}) \longmapsto (\mathfrak{S}, c^{\mathbf{QP}}, c^{\mathbf{QS}}).$$

Proof. By **QP-Phase**, p is constant on each scalar-redescription orbit in both variables. The universal property of the coequalizer therefore gives one and only one $\bar{p}$ with $\bar{p}(qx, qy) = p(x, y)$. If two quotient points have identical $\bar{p}$ profiles, **QP-Sep** says that their representatives lie in the same orbit; hence the quotient points are equal. Thus $\bar{p}$ separates the declared standing-relevant distinctions.

The determinate use of state presentations, comparisons, lawful transformations, continuation, and reusable predictions instantiates Admissibility, Standing, Reference, and Irreversibility. Retaining distinct standing for two representatives in one lawful scalar orbit would privilege a presentation without a changed p -witness, contradicting Reference and Admissibility. Identifying two different orbit classes would, by profile completeness, erase a registered transition distinction and contradict Standing. Therefore the standing relation is neither finer nor coarser than the orbit relation: it factors uniquely and exactly through q .

By **QP-Cont**, $q(C(\gamma x)) = q(Cx)$, so coequalizer universality gives the unique map $\bar{C}$ with $\bar{C}q = qC$; preservation of p gives preservation of $\bar{p}$. By **QP-Comp**, changing either representative by a scalar changes the product only within its scalar orbit. Hence $\boxtimes$ descends uniquely. Multiplicativity of p descends to $\bar{p}$, and the associator, unitors, symmetry, and monoidality of continuation descend because their representative diagrams commute.

For a lawful arrow T , equivariance gives the unique quotient arrow $\bar{T}(qx) = q'(Tx)$. Preservation of p , C , and $\boxtimes$ proves every certificate square for $\bar{T}$. The same universal-property argument after restriction gives $j^*\bar{p} = \bar{j}^*p$, $j^*\bar{C} = \bar{j}^*C$, and $j^*\boxtimes = \bar{j}^*\boxtimes$. Uniqueness makes identities and composition exact, proving the functor QClose_U and the naturality coordinate δ_{nat} . $\square$

Remark 10.3 (Positive-construction lock). Proposition 10.2 manufactures none of the raw certificate data. It chooses no presentation carrier, scalar action, transition weight, comparison class, continuation, record, clock, generator, regulator, ordering, or counterterm. It derives the standing quotient and the descended/natural structure only after those data and their six representative-level proofs are positively supplied.

10.3 The occupied WDW certificate and its comparison class

Let

$$\Phi_9 : \mathcal{H}_{\text{phys},9} \xrightarrow{\cong} \mathbb{C}$$

be the isometry in (37), put $\Omega_9 := \Phi_9^{-1}(1)$, and define

$$X_9 := \mathcal{H}_{\text{phys},9} \setminus \{0\}, \quad q_9 : X_9 \rightarrow X_9/\mathbb{C}^\times = \mathbb{P}\mathcal{H}_{\text{phys},9}.$$

On representatives set

$$p_9(u, v) := \frac{|\langle u, v \rangle_{\text{phys}}|^2}{\|u\|_{\text{phys}}^2 \|v\|_{\text{phys}}^2}.$$

Write $U_9(\tau, \sigma)$ for coefficient-preserving parallel transport, $D(H_9) = \mathcal{H}_{\text{phys},9}$ and $H_9 = 0$. Retain the separately typed source-attached record package $\mathcal{R}_9$, including V^{rec} , and the first-failure object Fail_9 .

Define the symmetric monoidal comparison groupoid $\mathcal{L}_9$ by

$$\text{Ob}(\mathcal{L}_9) = \mathbb{N}, \quad \mathcal{L}_9(n) = \mathcal{H}_{\text{phys},9}^{\widehat{\otimes} n}, \quad \mathcal{L}_9(0) = \mathbb{C},$$

with every unitary between the displayed one-dimensional Hilbert spaces as a morphism, tensor product given by addition of tensor powers, and the standard associator, unitors, and flip. If $\mu_n : \mathbb{C}^{\widehat{\otimes} n} \xrightarrow{\cong} \mathbb{C}$ is canonical multiplication, then

$$m_{9,n} := \Phi_9^{-1} \circ \mu_n \circ \Phi_9^{\widehat{\otimes} n} : \mathcal{L}_9(n) \xrightarrow{\cong} \mathcal{H}_{\text{phys},9}, \quad m_{9,n}(\Omega_9^{\widehat{\otimes} n}) = \Omega_9.$$

For independently specified $n, m \in \mathcal{L}_9$ and $\gamma, \gamma' \in \mathbb{C}^\times$,

$$(\gamma u) \widehat{\otimes} (\gamma' v) = (\gamma \gamma')(u \widehat{\otimes} v), \quad p_{9,n+m}(u \otimes v, u' \otimes v') = p_{9,n}(u, u') p_{9,m}(v, v'), \quad U_{9,n+m} = U_{9,n} \widehat{\otimes} U_{9,m}. \quad (43)$$

These data carry an explicit representative certificate c_9^{QP} . Positivity supplies ζ_{prob} ; scalar cancellation in the normalized ratio supplies ζ_{ph} ; the ordinary projective coequalizer supplies ζ_{ray} ; rank one makes equality of transition profiles equivalent to equality of orbits, supplying ζ_{sep} ; the first identity in (43) supplies independent $(\Gamma \times \Gamma)$ -equivariance through scalar multiplication, while the remaining identities and Hilbert coherence supply ζ_{comp} ; and unitary parallel transport supplies ζ_{cont} . Set

$$c_9^{\text{QS}} := \text{QClose}(c_9^{\text{QP}})$$

and let Ω_9^{WDW} denote the physical line, its raw certificate, the derived standing certificate, the represented algebra $p_9(\lambda I) = \lambda I$, the record, generator, and failure package.

Theorem 10.4 (W9.19 – certified standing closure and occupied export). *The theorem has two distinct conclusions.*

1. For every certified nondegenerate determinate admissible quantum predictive interior, W9.19 constructs the quotient-level state standing, separated transition weight, descended continuation, strong symmetric monoidal composition, and their arrow/restriction naturality:

$$(\mathfrak{S}, c_{\mathfrak{S}}^{\text{QP}}) \xrightarrow{\text{QClose}} (\mathfrak{S}, c_{\mathfrak{S}}^{\text{QP}}, c_{\mathfrak{S}}^{\text{QS}}).$$

None of these five δ -conclusions is a coordinate of $c_{\mathfrak{S}}^{\text{QP}}$.

2. The completed WDW-9 certificate positively supplies the raw certificate c_9^{QP} and hence occupies the derived model $\mathfrak{Q}_9^{\text{WDW}}$ with c_9^{QS} , nonzero positive physical line, faithful scalar algebra, reversible unitary continuation with $H_9 = 0$, coherent independent composition in $\mathcal{L}_9$, separate source-attached record, and retained first failure.

Consequently

$$\mathbf{QP}(\mathfrak{Q}_9^{\text{WDW}}; c_9^{\text{QP}}) \implies \mathbf{QS}(\mathfrak{Q}_9^{\text{WDW}}; c_9^{\text{QS}}), \quad H_9^{\text{QS-EXPORT}} = \text{PASS}.$$

This status is strictly downstream of and does not alter $G_9 = \text{PASS}$.

Proof. Part 1 is Proposition 10.2. For Part 2, W9.13 gives the nonzero positive quotient, isometry Φ_9 , and faithfully represented scalar algebra. The displayed calculations above verify every ζ -coordinate without assuming a δ -coordinate. WDW-7, imported through W9.14, proves unitary reversible parallel transport and the zero covariant Hamiltonian on the full one-dimensional fibre. W9.14 keeps the nonconstant record in the source lineage rather than treating it as a second physical state; W9.16 retains every first failure; and W9.17–W9.18 place the coordinates in one licensed certificate. Applying QClose supplies the derived certificate. No step uses a spatial Schrödinger carrier or its Hamiltonian classification. $\square$

Corollary 10.5 (Principal theorem). *The principal theorem stated in the front matter holds.*

Proof. Items 1–4 are Theorems 3.2, 4.7, 7.4 and 7.7. Items 5–7 are Theorems 8.1, 8.2 and 8.3. Item 8 is Theorem 10.4. The core and export statuses are Theorems 9.3 and 10.4, respectively. $\square$

10.4 Exact non-export ledger

Table 10: Boundary of the WDW-9 certified quantum-standing export.

Coordinate	Status
Representative QP data	Positively constructed for the accepted rank-one WDW line, with all six ζ -certificates explicit.
Standing quotient, descended transition/-continuation/composition, and naturality	Derived by W9.19 as the five δ -coordinates; none is assumed in QP .
Independent composition	Certified for every pair in $\mathcal{L}_9$; no arbitrary external system is adjoined.
Clock and record	Exported with separate types; the changing record is not a second physical state.
State plurality or a Hilbert carrier of dimension at least two	Not exported; $\mathbb{P}\mathcal{H}_{\text{phys},9}$ is a singleton.
A target-preserving carrier map $\mathcal{H}_{\text{phys},9} \cong \mathbb{C} \rightarrow L^2(\mathbb{R}^d)$	Not constructed or implied.
Flat space, Δ , V , $\hbar$, m , Wigner–Stone data, or SSQ_1	Not exported.
Measurement, open-system, field-theoretic, relativistic, ultraviolet, or full quantum-gravity dynamics	Not exported.

Handoff polarity. W9.19 is substantive but scoped. Its generic part converts certified representative data into a standing-bearing quotient and proves descent/naturality. Its WDW part supplies one occupied rank-one input and output. A downstream spatial theorem may import the generic quotient construction while bearing its own faithful spatial-projection burdens; it does not enlarge the occupied WDW carrier.

11. Dependency audit, gate matrix, and robustness analysis

11.1 Nineteen-node theorem chain

Table 11: Acyclic WDW-9 dependency chain.

Node	Burden	Antecedents	Closing result
W9.1	Exact WDW-8 handoff and target	W8.17	Theorem 1.2
W9.2	Thirteen typed fibres	W9.1; ARC target	Theorem 2.2
W9.3	Complete overlap graph	W9.2; WDW-1–8	Theorem 2.4
W9.4	Positive joint section	W9.3	Theorem 3.2
W9.5	Witness-reflecting completion/tensor colimit	W9.4; Module-D method	Theorem 4.3
W9.6	Complete certificate and typed joint probe	W9.4–5	Theorem 4.5
W9.7	Certificate and fibre skins	W9.6	Theorem 4.7
W9.8	Candidate grammar completeness	W9.2, W9.5–7; fixed-domain theorems	Theorem 5.3
W9.9	Staged fibre/tuple status and failure	W9.8	Theorem 5.6
W9.10	Candidate-specific response realization and thirteen typed fibre-finality results	W9.6–9; WDW-1–8	Equation (27); Theorem 6.16
W9.11	Certificate finality, compatibility descent, residue, assignments	W9.4, W9.7, W9.10	Proposition 7.2; Theorem 7.4
W9.12	Lawful automorphism orbit	W9.11	Theorem 7.7
W9.13	Physical joint kernel/quotient	W9.4, W9.10–12; WDW-6	Theorem 8.1
W9.14	Time, boundary, clock and record	W9.4, W9.13; WDW-3/7	Theorem 8.2
W9.15	Constraint-native WDW lineage	W9.13–14; WDW-1–8	Theorem 8.3
W9.16	No backflow and failure boundary	W9.1–15	Theorem 9.1
W9.17	Complete realization closure	W9.12–16	Theorem 9.2
W9.18	G_9 endpoint	W9.17	Theorem 9.3
W9.19	Certified standing quotient/descent and occupied export	W9.13–18; kernel necessity; explicit c^{QP}	Proposition 10.2; Theorem 10.4

No row cites a later row. In particular, the endpoint and orbit do not appear in the construction of the joint section, probe, or fibre residues, and the quantum-standing export does not appear in any antecedent of the core endpoint.

11.2 Gate matrix

Table 12: WDW-9 gate decisions.

Gate	Requirement	Pass evidence	Status
G9-01	Exact $I_{8 \rightarrow 9}$ import	Theorem 1.2; source-lineage table	PASS
G9-02	Role set fixed	Definition 2.1; Theorem 2.2	PASS
G9-03	Overlap graph complete	Lemma 2.3; Theorem 2.4	PASS
G9-04	Positive joint section	Lemma 3.1; Theorem 3.2	PASS
G9-05	Witness-reflecting completion	Equations (5)–(9); W9.5	PASS
G9-06	Certificate and joint probe	Equations (12)–(20); W9.6	PASS
G9-07	Maximal typed skins	Equations (21)–(22); W9.7	PASS
G9-08	Candidate grammar exhaustive	Lemma 5.2; Theorem 5.3	PASS

Gate	Requirement	Pass evidence	Status
G9-09	Eligibility/fidelity separated	Definition 5.4; staged-status theorem W9.9	PASS
G9-10	Compatibility/failure complete	Equation (24); first-failure theorem	PASS
G9-11	Thirteen typed fibre residues	Candidate-specific R_i tests; Equations (29)–(30); Propositions 6.3–6.15	PASS
G9-12	Compatibility descent/live residue	Lemma 7.1; Equations (34)–(35)	PASS
G9-13	Actual-certificate assignment orbit	Definition 7.5; Table 8; W9.12	PASS
G9-14	Physical joint kernel/quotient	Equation (37); Theorem 8.1	PASS
G9-15	Time/clock/boundary	Theorem 8.2	PASS
G9-16	WDW lineage	Equation (38); Theorem 8.3	PASS
G9-17	Failure/nonclaims pre-served	Equation (39); Theorem 9.1; Table 9	PASS
G9-18	Endpoint honestly reported	Theorems 9.2–9.3	PASS
H9-QS	Certified standing closure and occupied export	Proposition 10.2; explicit ζ/δ separation; Theorem 10.4	PASS

11.3 Adversarial objections and exact resolutions

The following analysis tests the proof at its load-bearing interfaces. An objection is resolved only by the cited construction; a label such as “kernel-faithful” or “complete” carries no independent proof force.

Table 13: Robustness objections A–BB.

ID	Objection	Exact resolution	Status
A	Thirteen nonempty fibres are treated as jointly nonempty.	W9.3 proves overlap completeness; Table 4 checks every edge; W9.4 exhibits $\Xi_9^* \in \mathfrak{J}_9$.	Closed
B	The tuple copies the desired conclusion.	Equation (4) uses independent predecessor occupants; compatibility is proved afterward and no later object occurs in its definition.	Closed
C	Roles were chosen after inspecting occupants.	The updated ARC and WDW-9 frozen contract fix R1–R13; W9.2 proves completeness from the target coordinate inventory before residue formation.	Closed
D	A standing-bearing overlap is omitted.	Lemma 2.3 inventories geometric, quantum, physical and recovery coordinates and proves factorization through E01–E20.	Closed
E	Arbitrary labels multiply residue capacity.	Residues inhabit typed carrier coproducts; eligibility depends on structural signatures and complete probes, not user-supplied labels.	Closed
F	Role tags are mistaken for a unique assignment.	All compatible injections are formed in (34); Lemma 7.3 and W9.11–12 compute the assignment and orbit.	Closed
G	Fibre singletons are multiplied without compatibility.	W9.4 establishes compatibility before W9.10; every $\mathcal{Q}_{9,i}$ is defined using occurrence in a compatible tuple.	Closed
H	The joint probe reads only headline outputs.	Θ_9^{cert} includes the tensor probe, thirteen augmented fibre probes, twenty incidence probes, and every assembly lock; Lemma 4.4 proves separation.	Closed
I	Skin erases a physical difference.	W9.7 proves maximality separately for S_9^{cert} and $S_{9,i}$; every listed physical coordinate factors through the corresponding probe.	Closed
J	Kernel necessity substitutes for population.	F1 supplies governance only; W9.4 independently exhibits nonvacuum, nonzero, changing-record and recovery occupants.	Closed

ID	Objection	Exact resolution	Status
K	WDW-8 recovery is identified with the WDW-6 ray.	E16 retains $\{1\}_{\text{phys}} \sqcup (0, \delta_0]_{\text{rec}}$ and the no-convergence statement; W9.13 uses only the physical anchor as state.	Closed
L	A WKB element is called a physical state.	Equation (37) is built from $F_A^{(0)}$; recovery elements remain comparison objects on the disjoint component.	Closed
M	WDW-7's zero line Hamiltonian contradicts recovery.	W9.14 types the physical line and recovery conditional equation separately; stationary ray and changing source record coexist.	Closed
N	WDW-4B populates the core gauge fibre.	F6 is the independently typed empty family; E06/E10 require absence of Branch-B data.	Closed
O	Empty Gauss is advertised as Standard-Model closure.	Proposition 6.8 proves only one empty-family class and expressly treats nonempty gauge structure as an enrichment.	Closed
P	Compact boundarylessness is globalized.	W9.14 says boundary time is absent on the core and retains boundary-generator targets as enrichments.	Closed
Q	Quotienting lapse/shift/foliation erases the physical clock.	The augmented F7/F12/F13 probes distinguish presentation variables from the function-valued clock and nonconstant record; only probe-preserving changes enter their skins.	Closed
R	The WDW equation is formal ADM substitution.	W9.15 traces carrier, differential, regulator, successor domain, anomaly closure, state, quotient, clock and recovery.	Closed
S	AASC alone fixes ordering or counterterms.	F9 imports the ordering, regulator grammar and finite profile as antecedents; W6R.11 derives the local coefficients, and W9.15 denies an AASC derivation.	Closed
T	Zero-theta Branch-A closure is generalized.	F9/F10, E13/E14, and Θ_9^{cert} retain branch, theta, counterterm, domain, endpoint, and exclusions.	Closed
U	The rank-one quotient is presented as all quantum gravity.	W9.13 and the boundary box restrict it to the pointed Branch-A target.	Closed
V	"Complete" smuggles global, UV, singular, asymptotic or SM claims.	Table 9 and Theorem 9.1 define certificate completeness and retain all richer-scope nonclaims.	Closed
W	WDW-9 repairs an upstream failure.	Proposition 1.3, the value-reflecting no-repair theorem, and W9.16 halt at the first failed gate.	Closed
X	Predecessor failures/nonclaims disappear.	They are coordinates of $I_{8 \rightarrow 9}$, $\widehat{W}_9$, the augmented and complete probes, Fail_9 , G_9 , and the report ledger.	Closed
Y	Equivalence is enlarged after seeing a rival.	Tensor, fibre, and complete probes are fixed before residue formation; W9.7 proves maximality on each stated domain.	Closed
Z	A plural family is hidden by stronger skin.	Fibre classes use $S_{9,i}$, compatibility descends before W9.11, and the live residue is the union over all compatible injections.	Closed
AA	The eight-fibre Module-D theorem is applied automatically to thirteen fibres.	W9.2–12 rebuild the schema, tensor ledger, complete certificate, both probe levels, typed skins, thirteen fibres, residue, and orbit.	Closed
AB	Equal stress tensors replace structured source ancestry.	E03/E04/E19/E20 are proved in the full source class with attachment certificate and witness; Section 3 states the common tuple.	Closed
AC	The automorphism group is declared trivial from names.	Table 8 gives pairwise nonisomorphic carrier/map/probe invariants; Proposition 7.6 proves every residue fixed.	Closed
AD	The colimit creates an undeclared tensor or identification.	Pointwise Kan formulas and the least generated relation prove exact generation; $\widehat{W}_9(b_e) = \mathfrak{B}_e$ makes unequal base values unidentifiable.	Closed
AE	Realization selection is promoted to decoherence, arrow or phenomenology.	Those claims are absent from the theorem and explicitly excluded in Table 9.	Closed

ID	Objection	Exact resolution	Status
AF	The capstone restates PASS labels.	W9.3–12 are new overlap, positive gluing, completion, probe, skin, exhaustion, fibre, residue and orbit theorems.	Closed
AG	A probe-identical rival is rejected by prose.	W9.7 puts certificate-probe-identical pairs in S_9^{cert} and fibre-probe-identical pairs in $S_{9,i}$; a missing probe factor would defeat W9.6.	Closed
AH	The singleton is vacuous or degenerate.	W9.4 explicitly contains the nonvacuum F3 history, nonzero F11 quotient, nonconstant F12 record and populated F13 recovery germ.	Closed
AI	A tensor-colimit element is silently treated as a thirteen-fibre certificate.	Equations (9) and (12) define distinct objects; the full probe is constructed only after the certificate projections.	Closed
AJ	Source locking alone is claimed to prevent quotient identification.	The proof uses the value-reflecting sorts $\widehat{W}_9(b_e) = \mathfrak{B}_e$; Proposition 4.2 explicitly notes that a singleton status witness would not suffice.	Closed
AK	A global skin relation is applied to fibre occupants outside its domain.	Equations (21)–(22) separate the domains; $\rho_i = [x_i^*]_{S_{9,i}}$ is formed only in Fi .	Closed
AL	Reflexivity makes LIVE and SKIN overlap as exclusive dispositions.	Definition 5.4 has four disjoint raw statuses; skin is a binary relation, and live is defined later only for quotient classes in (35).	Closed
AM	A colimit mediator $L \rightarrow P_9^{\text{ten}}$ is relabeled as a functor on ledger presentations.	Equations (15)–(17) construct the mediator groupoid, prove functoriality under ledger isomorphisms by colimit uniqueness, and use its discrete π_0 as the actual probe codomain.	Closed
AN	Fibre uniqueness is inserted into target fidelity.	Section 2 splits $\theta_i = (\alpha_i, \Omega_i)$ and excludes the canonical response value; TESTABLE status fixes only the antecedent. The broad $B_{9,i}$ and $Q_{9,i}^{\text{pre}}$ precede the independent 9C.i response theorem.	Closed
AO	Equal probe values are said to construct an accepted presentation arrow.	Lemma 4.4 and W9.7 assert only kernel-pair equality and projection. The deleted converse is not used; restricted quotients and compatibility descent operate directly on the relation.	Closed
AP	The set $\mathfrak{J}_9$ is later used as a groupoid without morphisms.	Equation (3) now defines the full compatible subgroupoid and specifies its restriction-preserving tuple arrows, making β_9 and all projections actual functors.	Closed
AQ	Fibre-local eligibility predicates are evaluated on the whole coproduct.	Equations (32)–(33) use the typed coproduct eliminator, evaluating the Fi predicate only on the Fi summand; structural signatures protect the carrier distinction.	Closed
AR	The probe ledger gives E01 the wrong fibre pair.	The overlap registry, manuscript and probe ledger all record E01 as F2–F3, and the publication check compares every incidence-probe domain with the overlap registry.	Closed
AS	Fibre kernel-pair replacement changes the tensor ledger because no fibre arrow is available.	The normalized interface datum D_9 contains exactly the probe and lock factors used by completion; Proposition 4.9 derives equal mediator classes by colimit uniqueness, and Proposition 7.2 then proves the survivor certificate quotient singleton without fibre arrows.	Closed
AT	Source-theorem applicability is treated as if it constrained the response carried by an arbitrary candidate.	Equation (27) introduces the candidate response $\omega_i(x)$ and requires $R_i(\alpha_i, \omega_i(x))$ before SURVIVES; each 9C.i proof applies its source theorem only after that bridge.	Closed
AU	The realization relation is canonical-output equality under another name.	Table 7 states only source equations, solution membership, universal properties and commuting diagrams; no row contains x_i^* , ω_i^* , $S_{9,i}$ or augmented-probe equality.	Closed
AV	A predecessor constructs the intended response but has not been shown to classify every response-law realizer.	Each 9C.i proof places the carried response in the predecessor’s quantified comparison, solution or universal-property domain; the cited uniqueness or exhaustion theorem then identifies its probe class.	Closed

ID	Objection	Exact resolution	Status
AW	A correct-input candidate with an altered response silently survives until the desired equality is asserted.	The ordered response disposition first checks $t_{i,k}$ and then every $r_{i,\ell}$; the least failed response law is recorded before $A_{9,i}$ is formed. The F9 proof displays the coefficient equation explicitly.	Closed
AX	Adding response-law tests could make the survivor fibres vacuous.	Every canonical predecessor construction satisfies its displayed R_i , and c_9^* gives each occupant a compatible extension; hence $x_i^* \in A_{9,i}$ before singleton cardinality is concluded.	Closed
AY	An Fi proof obtains its incident values by assuming finality of the fibre at the other end of an overlap.	Lemma 6.2 derives every $q_{e,i}$ value functorially from the Fi source comparison before tuple compatibility is used; no 9C.j premise occurs.	Closed
AZ	The response relations are new post hoc constructors added after inspecting rivals.	Table 7 extracts only equations, universal properties and diagrams from the accepted predecessor theorem domains, fixes them before testing $B_{9,i}$, and adds no W9.8 constructor.	Closed
BA	W9.19 assumes its projective-standing conclusion inside QP and then merely renames it QS .	Definition 10.1 places only representative data and six ζ -proofs in c^{QP} and expressly withholds standing descent, descended transition/continuation/composition, and naturality. Proposition 10.2 constructs the five δ -conclusions by coequalizer universality, kernel Standing/Reference, and descent.	Closed
BB	Membership in the export category follows from rank one or correctly typed fields without a predictive certificate.	QPred requires all six ζ -proofs and QStand additionally requires the five δ -proofs. Rank and types alone imply neither membership.	Closed

11.4 Publication verification

Finite consistency checks verify:

- the declared source lineage;
- 13 fibres and 20 generating overlap edges;
- 19 theorem nodes, 18 core PASS gates, and one downstream handoff PASS;
- 13 live residue classes and one compatible assignment orbit;
- 54 closed adversarial objections; and
- agreement of the endpoint payload with the printed theorem chain.

These checks supplement the proof. They do not certify the analytic, geometric, or category-theoretic theorems merely by finding expected strings.

Audit conclusion. The dependency graph is acyclic, the joint object is positively populated before completion, the tensor ledger is separated from the complete certificate, all probes and skins are typed on their actual domains, every carried response passes a candidate-specific realization relation before source uniqueness is invoked, compatibility descends before the live residue is formed, and no adversarial objection A–BB survives its cited theorem chain. The exact endpoint is the scoped $G_9 = \text{PASS}$ result of Theorem 9.3; the additive quantum-standing export is the separate $H_9^{\text{QS-EXPORT}} = \text{PASS}$ result of Theorem 10.4.

References and source lineage

References

- [1] A. J. Maley, *Non-Degenerate Construction and the Kernel of Admissibility: An Exhaustion Theorem for Standing, Reference, and Irreversibility*, 2026.
- [2] A. J. Maley, *Fixed-Domain Exhaustion and the Complete Disposition of Admissibility-Bearing Variations*, 2026.
- [3] A. J. Maley, *Same-Scope Operator Exhaustion under AASC*, 2026.
- [4] A. J. Maley, *Bivalence of Admissibility and Standing-Bearing Endpoint Status*, 2026.
- [5] A. J. Maley, *The Physical Clock as a Quotient*, 2026.
- [6] A. J. Maley, *Complete Certificate Closure under AASC: Integrated Modules A–D Closure*, 2026.
- [7] A. J. Maley, *Connection-Typed Canonical Projection of the Frozen Einstein–Source Realization*, WDW-1, 2026.
- [8] A. J. Maley, *Groupoid-Faithful Proper-Redescription Descent of the AASC Canonical Gravity Realization*, WDW-2, 2026.
- [9] A. J. Maley, *Normal Deformation as Constrained Refoliation: Groupoid-Faithful Hypersurface-Deformation Closure under AASC*, WDW-3, 2026.
- [10] A. J. Maley, *Classical Matter Occupancy and Joint-Constraint Closure under AASC*, WDW-4, 2026.
- [11] A. J. Maley, *Configuration-Space Differentiation and the Constraint-Native Wheeler–DeWitt Candidate Family*, WDW-5, 2026.
- [12] A. J. Maley, *Quantum Constraint Survival and Renormalized Physical Realization of the Wheeler–DeWitt Family*, WDW-6, 2026.
- [13] A. J. Maley, *Regular Phase-Jet Sufficiency and Canonical Rank-One Target Quotient Decision*, WDW-6R technical result, 2026.
- [14] A. J. Maley, *Profile-Clock Exhaustion and Constraint-Induced Cauchy-Germ Evolution*, WDW-7, 2026.
- [15] A. J. Maley, *Covariant–Canonical Reconciliation and Controlled Semiclassical Recovery*, WDW-8, 2026.
- [16] A. J. Maley, *Schrödinger Dynamics as the Unique Minimal First-Response Hamiltonian Projection of AASC Quantum Standing*, source-integrated handoff edition, Main Theorem 36.1, 6 August 2026.

Reference and source-lineage note

Table 1 identifies the declared theorem objects and their load-bearing uses. Source lineage protects the publication interfaces from accidental substitution; it is not used as a substitute for the proofs in this manuscript. The Schrödinger manuscript is a downstream comparison target for Section 10; no WDW-9 theorem imports its conclusions.